

Testing for Multiple-Horizon Predictability: Direct Regression Based versus Implication Based

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Research in finance and macroeconomics has routinely employed multiple horizons to test asset return predictability. In a simple predictive regression model, we find the popular scaled test can have zero power when the predictor is not sufficiently persistent. A new test based on implication of the short-run model is suggested and is shown to be uniformly more powerful than the scaled test. The new test can accommodate multiple predictors. Compared with various other widely used tests, simulation experiments demonstrate remarkable finite-sample performance. We reexamine the predictive ability of various popular predictors for aggregate equity premium. (*JEL* C32, C53, C58, G12, G14, G17)

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The predictability in asset returns is often examined beyond one forecast horizon. Fama and French (1988) and Campbell and Shiller (1988) found that accumulated short-run predictability can be the dominant influence on longer-run movements in asset prices. Ang and Bekaert (2007), Cochrane (2008), Lettau and van Nieuwerburgh (2008), Bansal and Shaliastovich (2013), and Rapach, Ringgenberg, and Zhou (2016) are recent studies of return predictability in which analyses over multiple horizons (or even at an infinite horizon) play a prominent role. Long horizon predictability tests are also useful in macroeconomics, for example, to detect predictable components in interest rates, inflation, and exchange rates.¹

Strong persistence of many useful predictors is a stylized fact in the return predictability literature. Unless the predictor is strictly exogenous, strong

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¹ See, for example, Mark (1995), Kilian (1999), Gourinchas and Rey (2007), Chen, Rogoff, and Rossi (2010), Colacito and Croce (2011), and Chen and Tsang (2013).

persistence can cause large size distortion to standard tests that are typically formed under the assumption of weak dependence. This issue has motivated several recent influential studies in the setting of the standard (or *short-run, single-horizon*) predictive regression.² In the *multiple-horizon* regression, our primary focus in this article, the outcome variable (e.g., multiple-period returns) is the aggregation of the single-horizon outcome. The potential effects of the predictor's persistence on inference are even further amplified because the outcome is overlapping, and thus also dependent. A large amount of empirical results favoring predictability over *longer horizons* are based on the test using the Newey-West (1987) (or Hansen-Hodrick (1980)) standard error. The Newey-West test can seriously overreject in finite samples due to the interaction between the persistent regressor and serially correlated errors. The standard error proposed by Hodrick (1992), section 1), which uses the moving-average structure of the aggregated error under the null hypothesis (of nonpredictability), performs better in finite samples (Ang and Bekaert 2007). However, the construction relies on weak dependence. Wei and Wright (2013) derive the asymptotic distribution of Hodrick's test (which is a nonstandard distribution) with strongly dependent regressors.

One approach that accommodates strongly persistent predictors is the scaled test, proposed by Valkanov (2003) and Hjalmarsson (2011). They use the usual t -statistic divided by the square root of the horizon (referred to as the scaled t -statistic) to reflect inflation of the standard error as the horizon increases, and the procedure has gained popularity among practitioners. However, the scale is motivated by strong persistence, and as we demonstrate, it tends to overestimate the standard error if the predictor is just weakly dependent. Using the asymptotics that allow for long horizons and various degrees of persistence, we show that under the null of nonpredictability, the scaled t -statistic of Hjalmarsson (2011) converges to zero if $h(1 - \rho) \rightarrow \infty$, where h is the forecast horizon and ρ is the autoregressive slope of the predictor. Consequently, under the alternative, the scaled t -statistic converges to zero if $n^{-1/2}h(1 - \rho)^{1/2} \rightarrow \infty$, where n is the sample size. This means that with a weakly dependent predictor, the test has zero power if the horizon increases faster than $n^{1/2}$.

The practical implication is that regardless of whether predictability is detected in the first few horizons, the significance (if any) of the scaled test will completely disappear as the horizon increases, if the predictor is not very strongly persistent. This is also the pattern we found in our empirical analysis.

In this paper, we pursue a different approach to infer about the multiple-horizon predictability slope *through the implication of the short-run model*. This method does not estimate the multiple-horizon regression directly, and

² See Stambaugh (1999), Lanne (2002), Amihud and Hurvich (2004), Torous Valkanov, and Yan (2004), Campbell and Yogo (2006), Jansson and Moreira (2006), Amihud et al. (2009), Phillips (2014), Elliott, Muller, and Watson (2015), Kostakis et al. (2015), Choi, Jacewitz, and Park (2016), Lee (2016), Xu (2017), Andersen and Varneskov (2018), Lee, Shi, and Gao (2018), and Xu and Guo (2019), among many others.

thus is not subject to issues raised by overlapping observations. The so-called “implied estimator” dates back at least to Campbell and Shiller (1988) and Hodrick (1992), section 3), and was more recently used by Cochrane (2008) and Lettau and van Nieuwerburgh (2008). Under the building block model of short-run predictive regression, we derive the asymptotic distribution of the implied test under both hypotheses of predictability and nonpredictability. The implementation of the test is based on the bias-corrected bootstrap, which accounts for persistence of the predictor. We also prove that the test is uniformly more powerful than the scaled test. In particular, the implied test statistic improves the divergence rate by $h(1-\rho)$ if $h(1-\rho) \rightarrow \infty$. The power improvement comes from two key elements in the construction of the implied test, a more efficient point estimator and an adaptive variance estimator under the alternative. The new implied test can easily extend to cover multiple predictors.

In the simulation experiments with empirically relevant designs, we confirm that the substantial power improvement of the implied test carries over to finite samples, and we provide evidence for its appealing performance over various other popular tests.

Recent work by Phillips and Lee (2013) and Kostakis et al. (2015, section 5) also propose robust tests, both based on the idea of constructing mildly persistent instrumental variables. Importantly, both studies take an indirect approach. Phillips and Lee (2013) estimate the backward multiple-horizon regression, in which the parameter of interest is different from ours and coincides only under nonpredictability. The test of Kostakis, Magdalinos, and Stamatiogiannis (2015) uses long-horizon returns as the dependent variable but the construction is built on a different estimand from ours.

Our approach thus complements theirs. An important benefit is that we focus directly on *the multiple-horizon slope* (as defined by θ_h in Equation (4)) of the predictive regression in which only the outcome is aggregated and the predictor remains the same with the usual short-run predictive regression. Thus, our approach is able to offer a confidence interval for the parameter of dramatic economic meaning, the multiple-horizon slope and, thereby, the multiple-horizon expected returns. For example, Rapach, Ringgenberg, and Zhou (2016) consider a mean-variance investor who rebalances his/her portfolio at the frequency lower than the data frequency (e.g., managing asset allocation quarterly ($h=3$) or annually ($h=12$), given monthly observations). Thus, the investor’s asset allocation strategy is based on multiple-horizon expected returns and variances. Moreover, our test requires no tuning parameter. By contrast, as Phillips (2014) notes, the instrumental-variable-based approach requires the user’s input of a key tuning parameter (which determines how mild the persistence of the instrumental variable is), but a satisfactory data-dependent selection rule is currently lacking.

Wei and Wright (2013) also emphasize the importance of focusing on the multiple-horizon slope. They propose running the reverse-regression and then

recovering the parameter of interest, the multiple-horizon slope, and the limit distribution is obtained through the delta-method. However, their test is valid only when the predictor is weakly dependent and thus does not offer a solution to robust inference to unknown persistence.

The implied test we suggest is related to that of Rossi (2007), who also proposes drawing inference based on the model-implied identification. Her confidence-interval approach relies on a monotonicity assumption, which is justified under the asymptotics in which the horizon is a fixed fraction of the sample size. Moreover, her analysis focused on local-to-unity specification (thus ruling out a weakly dependent predictor) and did not provide asymptotic power properties or offer guidance on a practitioner's choice among competing procedures.

1. Model and Estimand

Consider the predictive regression

$$y_t = \mu_y + \beta x_{t-1} + u_{0t}, \quad (1)$$

where the predictor x_t follows

$$x_t = \mu_x + \rho x_{t-1} + u_{xt}, \quad (2)$$

with the autoregressive (AR) slope ρ . The error vector $u_t = (u_{0t}, u_{xt})'$ is a bivariate martingale difference sequence (MDS) with the variance matrix $\text{Var}(u_t) = \Omega = \begin{pmatrix} \sigma_0^2 & \sigma_{0x} \\ \sigma_{0x} & \sigma_x^2 \end{pmatrix}$.³ The (contemporaneous) correlation $\delta = \sigma_{0x} \sigma_x^{-1} \sigma_0^{-1}$ between two errors is often referred to as the endogeneity parameter and is known to induce nonstandard inference of model (1) if x_t is strongly persistent (Campbell and Yogo 2006).⁴ We focus on the simple AR(1) model for the predictor to make the theory transparent.

We are interested in the parameter θ_h , given by

$$\theta_h = \sum_{j=0}^{h-1} \rho^j \beta, \quad (3)$$

where h is the forecast horizon. The parameter θ_h coincides with the estimand of the horizon- h regression (given by Equation (4)) under models (1) and (2), which is popular in empirical finance and macroeconomics. To show

³ We clarify a subtle difference between the data-generating process (DGP) and the (short-run) regressions we actually run, the latter of which are given by Equations (1) and (2). The DGP underlying the theoretical development is (1), together with $x_t = \mu_x + \tilde{x}_t$, where $\tilde{x}_t = \rho \tilde{x}_{t-1} + u_{xt}$. As we are primarily interested in regression slopes, we assume without loss of generality that $\mu_x = 0$.

⁴ See also Cavanagh, Elliott, and Stock (1995), Lanne (2002), and Torous Valkanov, and Yan (2004) for earlier theoretical results.

this, adding together Equation (1) for the outcomes for the next h periods, $y_t, y_{t+1}, \dots, y_{t+h-1}$, and using sequential substitution to maintain x_{t-1} as the regressor, we have

$$\sum_{j=0}^{h-1} y_{t+j} = \mu_h + x_{t-1} \theta_h + u_{0t}(h), \quad (4)$$

where θ_h is as in Equation (3), $\mu_h = h\mu_y$, and $u_{0t}(h)$ is the aggregated error given by

$$u_{0t}(h) = \sum_{j=0}^{h-1} u_{0,t+j} + \mathbb{I}_{\{h \geq 2\}} \left(\sum_{j=0}^{h-2} \varphi_j u_{x,t+j} \right) \beta, \quad (5)$$

with $\varphi_j = \sum_{i=0}^{h-2-j} \rho^i$. For example, if $h=3$, $u_{0t}(h) = u_{0t} + u_{0,t+1} + u_{0,t+2} + [(1+\rho)u_{xt} + u_{x,t+1}] \beta$. The dependent variable in regression (4) is often the (log) return of an asset held for h periods, and $x_{t-1} \theta_h$ is thus interpreted as the horizon- h expected return. We refer to θ_h as the horizon- h regression slope, or the long-horizon (LH) regression slope.

Valid inference of θ_h relies crucially on the persistence of the predictor, that is, how close ρ is to 1. To model the level of persistence, we assume

$$\rho = 1 + cn^{-r}, \quad (6)$$

where n is the sample size and $0 \leq r \leq 1$. This specification accommodates a wide range of persistence typically modeled in the literature, allowing for local-to-unity or strongly persistent ($r=1$; Campbell and Yogo 2006, among many others), nearly stationary ($0 < r < 1$; Park 2003b; Phillips and Magdalinos 2007; Phillips 2014; Kostakis, Magdalinos, and Stamatogiannis 2015), and more traditionally, stationary predictors ($r=0$). The constant c is such that $c \in \mathbb{R}$ if $r=1$, $c < 0$ if $0 < r < 1$, and $-2 < c < 0$ if $r=0$.

Throughout the paper, we assume the horizon $h \rightarrow \infty$ but with the rate slower than the sample size n , that is,

$$h^{-1} + hn^{-1} \rightarrow 0. \quad (7)$$

The assumption $h \rightarrow \infty$ is used often in the literature to model long horizons, and it is particularly convenient in our setting when analyzing the power properties at long horizons (see Sections 2 and 3). The assumption $h/n \rightarrow 0$ is motivated by the practical convenience of the asymptotic distribution⁵ and the reasonable approximation of the finite-sample distribution (see Section 5). Sometimes we write $h = c_h n^{r_h}$, where $c_h > 0$ and r_h denotes the increasing rate of h , with $0 < r_h < 1$. We exclude $r_h = 1$, in which case the first term of the error $u_{0t}(h)$ (as in (5)) would behave like a unit root process, and thus the

⁵ That is, the nonstandard asymptotic distribution of test statistics considered subsequently (Theorems 1 and 3, when $r=1$) is the same as for inference of short-run predictability ($h=1$). See also the next footnote.

long-horizon regression (4) would be subject to issues of spurious regression (Phillips 1986) (e.g., the slope estimator is not consistent).⁶

A larger h in general leads to a lower quality of the least squares estimator θ_h in the regression (4), denoted by $\tilde{\theta}_h$, because it induces an increase of the marginal variance *and* the overlap of the residual $u_{0t}(h)$.⁷ For example, if $\beta=0$, the convergence rate of $\tilde{\theta}_h$ is $n^{[1+r-r_h-\min(r,r_h)]/2}$ (Theorem A1 in Appendix B); r_h enters the rate negatively. This rate reduces to the familiar $\sqrt{n}$ -rate as in the classical time-series regression (i.e., the predictor is stationary and weakly dependent $r=0$, and the horizon is fixed $r_h=0$). A similar effect also occurs for an alternative estimator of θ_h , shown in Section 3.

We focus on testing the null hypothesis $H_0: \theta_h=0$, or equivalently, $H_0: \beta=0$ (nonpredictability) if there is only one predictor. In empirical finance literature, the hypothesis H_0 is often examined by running multiple-horizon regressions for a set of horizons. The claim of predictive (in)ability of a certain predictor is reinforced if results confirm across horizons. One of most influential puzzles in the empirical asset pricing literature is that evidence of predictability tends to be stronger at a longer horizon for a persistent predictor (Campbell and Shiller 1988; Fama and French 1988).⁸

Throughout the paper, we impose the following assumption on the error process u_t :

Assumption 1. u_t is stationary and ergodic martingale difference sequence with $E(u_t^2|u_s, s \leq t-1) = \Omega$. x_0 is independent of u_t for $t \geq 1$ and satisfies $Ex_0^2 = o(n^{1/2})$.

Remark 1 (R -square). Under models (1) and (2), we can show that the population R^2 approaches 1 if $r_h < r$ (the not-long-enough horizon is dominated by the persistence of the predictor) and approaches 0 if $r_h > r$.⁹ This explains the hump shape of R^2 (as a function of the horizon) commonly observed in empirical studies for fairly persistent predictors. The horizon at which the R^2

⁶ The case $r_h=1$ corresponds to the asymptotics (originated in Richardson and Stock 1989; see also Valkanov 2003; Rossi 2007, among others) in which h is assumed to increase as a fixed fraction of n . We adopt the asymptotics based on (7) mainly for two reasons. First, given the spurious regression issues mentioned, the fixed-fraction asymptotics typically delivers tests (e.g., those based on T_{Scaled} and T_{IM} below) that are inconsistent under the alternative. Second, the fixed-fraction asymptotics is implementationally less convenient (e.g., for Bonferroni-type approach), because the limit distribution needs to be simulated for each h (Valkanov 2003).

⁷ For the increasing rate of the marginal variance, if $\beta=0$, $\text{Var}(u_{0t}(h)) \sim h$. If $\beta \neq 0$, the order of $\text{Var}(u_{0t}(h))$ depends on the persistence of x_t , ranging from h (when $r=0$) to h^3 (when $r=1$).

⁸ In contrast with (4), Bandi and Perron (2008) consider a different type of long-horizon regression in which $\sum_{i=1}^h x_{t-i}$ serves as the regressor in place of x_{t-1} . Using the market variance as the predictor, they find that such a forward-backward regression (with the Newey-West standard error) produces strong evidence for return predictability as the horizon increases (see also Sizova 2013; Bandi et al. 2019).

⁹ In fact, the explained variance in regression (4) (the variance of expected returns) is of order $n^{r+2\min(r_h,r)}$, and the residual variance (the variance of unexpected returns) is of order $n^{h+2\min(r_h,r)}$. This can be shown by Lemma 1 (Appendix C, Equations (A2) and (A3)).

reaches the peak depends on the persistence of the predictor. For a strongly persistent predictor (e.g., the dividend-to-price ratio), the R^2 keeps increasing until a quite large horizon, while for a weakly dependent predictor, the R^2 may follow a declining shape for almost the entire range of horizons (e.g., the corporate-government bond returns spread (DFR), Figure 20d).

Remark 2 (The estimand). Although the asymptotic theory for inference we develop below hinges on Assumption 1, θ_h (as defined in (3)) is still the least-squares estimand of the multiple-horizon regression (4) under the maintained hypothesis in broader circumstance (e.g., allowing for serial correlation in u_{xt} , under the condition $r > 0$ and a certain fairly weak restriction on the horizon, in view of the results in Magdalinos and Phillips (2009, theorem 3.1)).

2. Long-Horizon Regression-Based Test

Data are $\{x_t, y_t : t = 1, \dots, n+h-1\}$. The long-horizon (LH) regression-based estimator of θ_h in (4) is written as

$$\tilde{\theta}_h = \left(\sum_{t=2}^n \tilde{x}_{t-1}^2 \right)^{-1} \sum_{t=2}^n \tilde{x}_{t-1} \sum_{j=0}^{h-1} y_{t+j},$$

where $\tilde{x}_{t-1} = x_{t-1} - (n-1)^{-1} \sum_{t=2}^n x_{t-1}$.

A popular test of H_0 based on the long-horizon regression is the scaled test (Valkanov 2003; Hjalmarsen 2011). The test is built on the usual t -statistic scaled by $h^{-1/2}$,

$$\mathcal{T}_{\text{Scaled}}(\theta_{h0}) = v_{\text{Scaled}}^{-1}(\tilde{\theta}_h - \theta_{h0}),$$

for a generic θ_{h0} , where $v_{\text{Scaled}}^2 = h \left(n^{-1} \sum_{t=2}^n \hat{u}_{0t}^2(h) \right) \left(\sum_{t=2}^n \tilde{x}_{t-1}^2 \right)^{-1}$ and $\hat{u}_{0t}(h)$ is the ordinary least squares (OLS) residual from (4).¹⁰ The scale $h^{-1/2}$ was originally motivated by the belief that the predictor is strongly persistent (local-to-unity). Remark 6 explains why the scale works for a boarder class of predictors (as long as the horizon is short relative to the predictor persistence) and also delineates the cases when the scale fails (inducing variance overestimation).

We establish the asymptotic behavior of $\mathcal{T}_{\text{Scaled}}(0)$ in Theorem 1 for the full range of persistence levels of the predictor. The results are based on the asymptotic distribution of $\tilde{\theta}_h$, which is provided in Theorem A1 in Appendix B.

To fix notations, let $\mathcal{Z} \sim \mathcal{N}(0, 1)$ be a standard normal variable and $B_x(s)$ be a standard Brownian motion process that is independent of $\mathcal{Z}$.

¹⁰ We use the scaled t -statistic proposed by Hjalmarsen (2011). The recommended test statistic in Valkanov (2003) uses the scale $n^{-1/2}$ and has the limit distribution different from that of Hjalmarsen's t -statistic, when the AR root is local to unity. Valkanov's result is based on the asymptotics in which h increases at the same rate as n ; see discussions after Equation (7).

Denote $J_{x,c}(r) = \int_0^r e^{(r-s)c} dB_x(s)$ and $\tilde{J}_{x,c}(r) = J_{x,c}(r) - \int_0^1 J_{x,c}(s) ds$. Let $\tau_2(c) = (\int \tilde{J}_{x,c}^2)^{-1/2} \int \tilde{J}_{x,c} dB_x$ and $\tau_1(c, \delta) = \delta \tau_2(c) + (1 - \delta^2)^{1/2} \mathcal{Z}$.

Theorem 1. Suppose that Assumption 1 holds and $\beta = 0$. Under (6) and (7), the following results hold.

- (i) If $r = 1$, $\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{d} \tau_1(c, \delta)$.
- (ii) If $0 < r_h \leq r < 1$, $\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{d} \mathcal{N}(0, 1)$.
- (iii) If $0 < r < r_h < 1$, $n^{-r/2} h^{1/2} \mathcal{T}_{\text{Scaled}}(0) \xrightarrow{d} \mathcal{N}(0, 2(-c)^{-1})$.
- (iv) If $0 = r < r_h < 1$, $h^{1/2} \mathcal{T}_{\text{Scaled}}(0) \xrightarrow{d} \mathcal{N}(0, (1 - \rho)^{-1}(3 - \rho))$.

Remark 3 (Bonferroni correction). Hjalmarrsson (2011) established Theorem 1 (i) for a fixed h . Theorem 1 (i) shows that when the predictor is local-to-unity, $\mathcal{T}_{\text{Scaled}}(0)$ has the usual limit distribution as in the short-run predictive regression (Cavanagh, Elliott, and Stock 1995; Campbell and Yogo (2006)), when the horizon is allowed to grow but with a rate slower than n . Thus, a Bonferroni test can be constructed, in the most conservative way, using critical values of $\tau_1(c, \hat{\delta})$ (where $\hat{\delta}$ is a consistent estimator of δ), over a preliminary confidence interval for the unknown constant c . This forms a standard practice when inferring about long-run stock return predictability (Valkanov 2003; Hjalmarrsson 2011).¹¹

Remark 4 (Robust scaled t -test). A common way to construct the confidence interval of c is to invert a unit root statistic, as suggested in Stock (1991). Phillips (2014) shows that such formed confidence interval is not informative (instead, it can be infinitely long) when $r < 1$, especially when $r = 0$, the traditional assumption of stationarity. Phillips thus suggests, with a focus on short-run predictability parameter, using instead the uniform confidence interval of Mikusheva (2007) that reaches the claimed coverage for all r such that $0 \leq r \leq 1$. Phillips's proposal extends to the current setting, and the associated Bonferroni procedure will be referred to as the *robust* scaled t -test.

Remark 5 (Potentially zero null rejection rate). As shown in Theorem 1 (i) and (ii), the robust scaled test provides advertised (but conservative) inference (by noting that $\tau_1(c, \delta) \xrightarrow{d} \mathcal{N}(0, 1)$ as $c \rightarrow -\infty$), conditional on the horizon increasing with a rate no faster than the rate at which the autoregressive root of the predictor approaches unity (i.e., $h(1 - \rho) \rightarrow 0$). If the horizon increases faster to exceed this benchmark, $\mathcal{T}_{\text{Scaled}}(0)$ shrinks to zero at the rate $n^{(r-r_h)/2}$, as Theorem 1 (iii) and (iv) indicates. When this happens, even with the right confidence interval of c , the robust scaled test rejects the null at any significance level with zero probability asymptotically (i.e., being *overly conservative*).

¹¹ Hjalmarrsson (2011) also proposes the multiple-horizon version of the Q-test (Campbell and Yogo 2006), which corrects for contemporaneous endogeneity. Section 5 examines the test's finite-sample properties. See also Phillips (2014) for a potential pitfall of using the Q-test.

Remark 6 (Variance overestimation). To understand Theorem 1(ii)-(iv), when $0 \leq r < 1$, $\mathcal{T}_{\text{Scaled}}(0)$ uses the variance of $\tilde{x}_{t-1}u_{0t}(h)$, multiplied by h , to approximate the long-run variance of $\tilde{x}_{t-1}u_{0t}(h)$, $\text{Var}(n^{-1/2} \sum_{t=2}^n \tilde{x}_{t-1}u_{0t}(h))$, under H_0 . This is valid when $r_h \leq r$, because both $h\text{Var}(\tilde{x}_{t-1}u_{0t}(h))$ and $\text{Var}(n^{-1/2} \sum_{t=2}^n \tilde{x}_{t-1}u_{0t}(h))$ have the leading term $h^2 n^r (-2c)^{-1} \sigma_x^2 \sigma_0^2$ (see the proof of Theorem A1). However, if $r_h > r$, $h\text{Var}(\tilde{x}_{t-1}u_{0t}(h)) \sim h^2 n^r$, which is of a larger order than $\text{Var}(n^{-1/2} \sum_{t=2}^n \tilde{x}_{t-1}u_{0t}(h)) \sim hn^{2r}$; that is, the actual variance is overestimated by the method of scaling. This leads $\mathcal{T}_{\text{Scaled}}(0)$ to shrink to zero at the rate $n^{(r-r_h)/2}$ when $r_h > r$.

The major cost of a severely undersized test is the potential loss of power. The following theorem accounts for the asymptotic properties of $\mathcal{T}_{\text{Scaled}}(0)$ under the alternative.

Theorem 2. Suppose that Assumption 1 holds and $\beta \neq 0$. Under (6) and (7), the following results hold.

- (i) If $r = 1$, $n^{-1}h\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{d} 3^{1/2}(\int \tilde{J}_{x,c}^2)^{1/2}$.
- (ii) If $0 < r_h < r < 1$, $n^{-(1+r)/2}h\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{p} (3/2)^{1/2}(-c)^{-1/2}$.
- (iii) If $r_h > r > 0$, $n^{-(1+r)/2}h\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{p} (-2c)^{-1/2}$.
- (iv) If $r_h > r = 0$, $n^{-1/2}h\mathcal{T}_{\text{Scaled}}(0) \xrightarrow{p} [\sigma_0^2 \sigma_x^{-2} (1-\rho)^2 + 2(1-\rho)\sigma_{0x} \sigma_x^{-2} \beta + \beta^2]^{-1/2} (1-\rho^2)^{-1/2} \beta$.

Remark 7 (Power loss). Theorem 2 implies that the power of the test $\mathcal{T}_{\text{Scaled}}(0)$ in general decreases with h eventually, though the test may have larger power than a short-run test when the range of h covers just a few periods, both outcomes that have been documented in the empirical literature.¹² For a fixed $\beta \neq 0$, $\mathcal{T}_{\text{Scaled}}(0)$ is inconsistent when $n^{(1+r)/2}h^{-1} \rightarrow C$, where $C \geq 0$ is a constant. In other words, a long horizon would cause zero power (and therefore inconsistency) of the scaled test. The threshold for such a “too long” horizon depends on the persistence of the predictor, and it is lower for a less persistent predictor. When the predictor is weakly dependent ($r=0$), a horizon with the rate $n^{1/2}$ would lead $\mathcal{T}_{\text{Scaled}}(0)$ to have zero power against the predictability hypothesis.

3. Test Based on the Implied Estimator

In this section, we propose an alternative procedure based on a different estimator of θ_h . The estimator, termed the *implied* estimator in this paper ($\hat{\theta}_h$ in Equation (8)), uses the implication of the short-run model into the long run, as in (3), and it has been used by several scholars (e.g., Cochrane 2008;

¹² Instead of directly assessing the divergence rate of the statistic under the alternative, Campbell (2001) considers a different approach for evaluating the power of the test.

Lettau and van Nieuwerburgh 2008). However, the asymptotic properties under a general setup, the efficiency over the LH regression-based estimator, and the associated (improved) inference (discussed subsequently) are new.

We define the implied estimator of θ_h as

$$\hat{\theta}_h = \sum_{j=0}^{h-1} \hat{\rho}^j \hat{\beta}, \quad (8)$$

where $\hat{\beta}$ and $\hat{\rho}$ are OLS estimators in the regressions (1) and (2). To construct a test statistic, we further define $q = (q_1, q_2)$, where

$$q_1 = \sum_{j=0}^{h-1} \rho^j, \quad q_2 = \sum_{j=0}^{h-1} j \rho^{j-1} \beta, \quad (9)$$

and $\hat{q} = (\hat{q}_1, \hat{q}_2)$, which is obtained by inserting $\hat{\beta}$ and $\hat{\rho}$ in (9). We denote $\hat{\Omega} = n^{-1} \sum_{t=2}^n \hat{u}_t \hat{u}_t'$, where $\hat{u}_t = (\hat{u}_{0t}, \hat{u}_{xt})'$ is the OLS residual in (1) and (2). The implied t -statistic is defined as

$$\mathcal{T}_{\text{IM}}(\theta_{h0}) = v_{\text{IM}}^{-1}(\hat{\theta}_h - \theta_{h0}), \quad (10)$$

where $v_{\text{IM}}^2 = \hat{q}' \hat{\Omega} \left(\sum_{t=2}^n \tilde{x}_{t-1}^2 \right)^{-1} \hat{q}$. Note that $\mathcal{T}_{\text{IM}}(\theta_{h0})$ is identical to $\mathcal{T}_{\text{Scaled}}(\theta_{h0})$ when $h=1$. Theorems 3 and 4 give the asymptotics of $\mathcal{T}_{\text{IM}}(0)$ under the null and alternative hypotheses, for h satisfying (7). The results are based on the asymptotic distribution of $\hat{\theta}_h$, which is provided in Theorem A2 in Appendix B.

Theorem 3. Suppose that Assumption 1 holds and $\beta=0$. Under (6) and (7), the following results hold. (i) If $r=1$, $\mathcal{T}_{\text{IM}}(0) \xrightarrow{d} \tau_1(c, \delta)$. (ii) If $0 \leq r < 1$, $\mathcal{T}_{\text{IM}}(0) \xrightarrow{d} \mathcal{N}(0, 1)$.

Theorem 4. Suppose that Assumption 1 holds and $\beta \neq 0$. Under (6) and (7), the following results hold.

- (i) If $r=1$, $n^{-1} h \mathcal{T}_{\text{IM}}(0) \xrightarrow{d} 2(\int \tilde{J}_{x,c}^2)^{1/2}$.
- (ii) If $0 < r_h < r < 1$, $n^{-(1+r)/2} h \mathcal{T}_{\text{IM}}(0) \xrightarrow{P} 2^{1/2}(-c)^{-1/2}$.
- (iii) If $r_h > r > 0$, $n^{-(1-r)/2} \mathcal{T}_{\text{IM}}(0) \xrightarrow{P} 2^{-1/2}(-c)^{1/2}$.
- (iv) If $r_h > r=0$, $n^{-1/2} \mathcal{T}_{\text{IM}}(0) \xrightarrow{P} (1-\rho^2)^{-1/2} [\sigma_0^2 \sigma_x^{-2} + 2(1-\rho)^{-1} \beta \sigma_{0x} \sigma_x^{-2} + (1-\rho)^{-2} \beta^2]^{-1/2} \beta$.

Remark 8 (The $\mathcal{T}_{\text{IM}}$ test in action). The implied test is based on the t -statistic $\mathcal{T}_{\text{IM}}(0)$, and critical values can be constructed in two ways. One is based on the Bonferroni procedure. In this approach, critical values are obtained as quantiles of $\tau_1(c, \hat{\delta})$ over a uniform Mikusheva-Phillips confidence interval for c , similar to the robust scaled test based on $\mathcal{T}_{\text{Scaled}}(0)$ (see Remark 4). Unlike $\mathcal{T}_{\text{Scaled}}(0)$, $\mathcal{T}_{\text{IM}}(0)$ does not vanish for a large h , for all r such that $0 \leq r \leq 1$. This generates

significant power improvement of the $\mathcal{T}_{\text{IM}}$ test over the scaled test (see Remarks 9 and 10). However, this approach is still conservative because of the Bonferroni step. Moreover, the Bonferroni approach is highly computationally demanding in the multiple-predictor model (Section 4). The other way of constructing critical values for $\mathcal{T}_{\text{IM}}(0)$ is to use the bootstrap. We consider the bias-corrected wild bootstrap (residual-based with the recursive design), and the detailed algorithm for implementation is given in Appendix A (Algorithm 1 and Remark A1).¹³ In Section 5.2, we discuss the finite-sample properties of these two approaches. We recommend to use the bootstrap implied test. The program codes in MATLAB are available as a supplement to the paper.¹⁴

Remark 9 (Power improvement of the $\mathcal{T}_{\text{IM}}$ test). If $\beta \neq 0$, the $\mathcal{T}_{\text{IM}}(0)$ statistic diverges at the rate $n^{(1+r)/2 - \min(r, r_h)}$, while $\mathcal{T}_{\text{Scaled}}(0)$ diverges at the rate $n^{(1+r)/2 - r_h}$, as indicated by Theorems 2 and 4. The rates, as Figure 1 illustrates, are a function of r and r_h . When $r_h > r$, $\mathcal{T}_{\text{IM}}(0)$ diverges with a faster rate than $\mathcal{T}_{\text{Scaled}}(0)$, by a margin of $n^{r_h - r}$. When $r_h < r$, $\mathcal{T}_{\text{IM}}(0)$ diverges as fast as $\mathcal{T}_{\text{Scaled}}(0)$, but with a scale constant in the limit approximately 15.5% (i.e., $\sqrt{4/3} - 1$) larger than that of $\mathcal{T}_{\text{Scaled}}(0)$.

Remark 10 (Sources of power improvement). The lower power of $\mathcal{T}_{\text{Scaled}}(0)$, than that of $\mathcal{T}_{\text{IM}}(0)$, comes from two sources: a less efficient point estimator and inconsistent variance estimation under the alternative. On the one hand, if $r_h < r$, $\hat{\theta}_h$ and $\tilde{\theta}_h$ are asymptotically equivalent (have the same convergence rate and asymptotic distribution; see Theorems A1 and A2, Appendix B), so only the second source contributes. For example, the asymptotic variance is $\text{Avar}(\hat{\theta}_h) = (-c)\beta^2/2$ (if $0 < r_h < r < 1$), while the variance estimator converges to $2(-c)\beta^2/3$. On the other hand, if $r_h > r \geq 0$, $\hat{\theta}_h$ converges faster than $\tilde{\theta}_h$, with the rate $n^{(1-3r)/2}$ (against almost $n^{(1-2r-r_h)/2}$, the convergence rate of $\tilde{\theta}_h$). Thus, both sources contribute to the power disadvantage of $\mathcal{T}_{\text{Scaled}}(0)$, and each of them reduces the divergence rate by $n^{(r_h-r)/2}$.¹⁵

Remark 11 (VAR approach). The construction of the estimator $\hat{\theta}_h$ and the test statistic $\mathcal{T}_{\text{IM}}(\theta_{h0})$ is closely related to what is referred to in the literature as the vector autoregression (VAR) approach (Hodrick 1992, section 3). Here, we drop the lagged return in the system of equations to obtain sharper inference

¹³ The author is grateful to a referee offering multiple suggestions that lead to the recommended bootstrap test.

¹⁴ Interested readers may download the codes, together with data for demonstration, from the author's Web site (sites.google.com/site/xukeli2015/research).

¹⁵ This can be shown using the intuition behind the diminishing size result of $\mathcal{T}_{\text{Scaled}}(0)$ when $r_h > r$ (Theorem 1(iii) and Remark 6). When $\beta \neq 0$, if $r_h > r$, the variance estimator that $\mathcal{T}_{\text{Scaled}}(0)$ uses, that is, $h \text{Var}(\tilde{x}_{t-1} u_{0t}(h)) \sim h^2 n^{3r}$, is of a larger order than $\text{Var}(n^{-1/2} \sum_{t=2}^n \tilde{x}_{t-1} u_{0t}(h)) \sim h n^{4r}$. If $\mathcal{T}_{\text{Scaled}}(0)$ used the variance estimator that consistently estimates $\text{Var}(n^{-1/2} \sum_{t=2}^n \tilde{x}_{t-1} u_{0t}(h))$, the divergence rate of $\mathcal{T}_{\text{Scaled}}(0)$ would be improved by $n^{(r_h-r)/2}$.

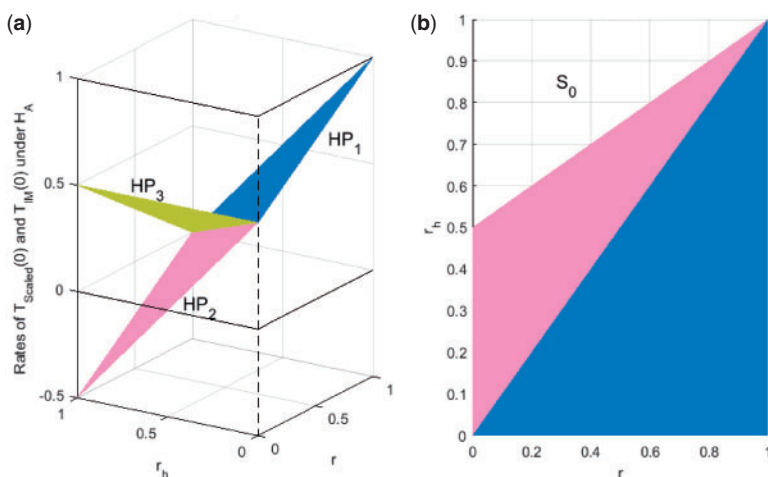


Figure 1

Divergence rates of $\mathcal{T}_{Scaled}(0)$ and $\mathcal{T}_{IM}(0)$ under the alternative hypothesis

In panel a, three triangular-shaped hyperplanes HP_1 , HP_2 , and HP_3 are shown, where HP_1 and HP_2 belong to one hyperplane. The divergence rate (i.e., d , if a statistic $\mathcal{T}$ diverges at the rate n^d) of $\mathcal{T}_{Scaled}$ is shown as $HP_1 \cup HP_2$, whereas the divergence rate of $\mathcal{T}_{IM}$ is shown as $HP_1 \cup HP_3$. Panel b is the projection of the 3D plot in panel a onto the (r, r_h) -plane with the vertical coordinate zero. $\mathcal{T}_{IM}$ is consistent within the entire unit square region, whereas $\mathcal{T}_{Scaled}$ is *inconsistent* in the region S_0 .

(i.e., assuming market efficiency in the weak form, but not in the semistrong form). This simplification was also used by Cochrane (2008) and Lettau and van Nieuwerburgh (2008) to construct the multiple-horizon parameter estimator.

Remark 12 (Conditional heteroscedasticity). The results in Theorem 3 are still valid under conditional heteroscedasticity for the persistence level $0 < r \leq 1$. The Online Appendix (available at the author's website) provides the proof in such cases by relaxing Assumption 1 so that conditional heteroscedasticity in errors is allowed. To cover the full range of persistence levels considered in this article (including the case $r=0$), the researcher needs to modify the $\mathcal{T}_{IM}(\theta_{h0})$ statistic by using White's standard error.¹⁶ As in this article, we mainly focus on the comparison with the scaled test, which was originally constructed using the usual standard error (Valkanov 2003; Hjalmarsen 2011), we call for future work to provide such an extension, together with theoretical justification and finite-sample assessment.

¹⁶ Changes also need to be made when constructing critical values for $\mathcal{T}_{IM}$. For the Bonferroni method, the conditional-heteroscedasticity-robust confidence interval for c , based on the work of Andrews and Guggenberger (2014), should be used. For the residual-based bootstrap, the wild bootstrap resampling scheme should be used, which we have incorporated in our implementation of the implied test.

4. Multiple Predictors

Next, we extend the implied test to multiple predictors x_t , which is $k \times 1$. While a wealth of the empirical literature of return predictability focuses on a single predictor, using multiple predictors is a natural way to combine information, especially given that many useful predictors have been suggested recently. Ang and Bekaert (2007) propose a present-value model that suggests that dividend yields and short rates predict excess returns. The model of Ang and Bekaert was used in Rapach, Strauss, and Zhou (2013). Santos and Veronesi (2006) and Amihud, Hurvich, and Wang (2009) used a two-predictor model with dividend yields, together with labor-income-consumption ratios.

To allow multiple predictors, write the system as

$$y_t = \mu_y + x'_{t-1}\beta + u_{0t}, \quad (11)$$

$$x_t = \mu_x + \rho x_{t-1} + u_{xt}, \quad (12)$$

where ρ is the $k \times k$ dimensional vector autoregression (VAR(1)) slope matrix, and the error $u_t = (u_{0t}, u'_{xt})'$ is a $(k+1)$ -dimensional martingale difference

sequence with the variance matrix $Var(u_t) = \Omega = \begin{pmatrix} \sigma_0^2 & \sigma'_{0x} \\ 1 \times 1 & 1 \times k \\ \sigma_{0x} & \Omega_x \\ k \times 1 & k \times k \end{pmatrix}$. Let $\Omega^{1/2}$ be

such that $\Omega = \Omega^{1/2}(\Omega^{1/2})'$, and write $\Omega^{-1/2} = (\Omega^{1/2})^{-1}$. In this section, we use the same notations as in the previous sections, and when $k=1$, these notations reduce to those we have seen earlier in this article.

Under models (11) and (12), using the recursive substitution, the horizon- h predictability parameter ($k \times 1$) can be written as

$$\theta_h = \sum_{j=0}^{h-1} (\rho')^j \beta. \quad (13)$$

The long-horizon regression estimator is obtained by running the OLS regression of $\sum_{j=0}^{h-1} y_{t+j}$ on the vector x_{t-1} . As an alternative, the implied estimator is obtained as $\hat{\theta}_h = \sum_{j=0}^{h-1} \hat{\rho}'^j \hat{\beta}$, where $\hat{\beta}$ is the OLS slope of (11), and $\hat{\rho}$ is obtained by the OLS VAR(1) regression of (12).

We are then interested in testing the hypothesis of d_R linear restrictions, $H_0: R\theta_h = R_0$, where R is a known $d_R \times k$ full-rank matrix, R_0 is a known $d_R \times 1$ vector, and $d_R \leq k$. Applications can be testing, over the next h periods, whether the predictors contained in x_{t-1} can *jointly* predict the return, or whether a particular predictor has predicting power once certain other predictors are included in the predictive regression.¹⁷

¹⁷ Note that if $d_R < k$, at a multiple horizon ($h > 1$), the null hypothesis here is generally not informative about the value of β even it can identify some values in θ_h , unless further assumptions are imposed (like assuming the coefficient matrix ρ to be diagonal); for example, a zero component of θ_h does not imply any component of β to be zero. See Xu (2019) for further discussions.

We denote $\phi = (\beta', (\text{vec}(\rho'))')$, where vec is the vectorization operator. To form the Wald statistic through the delta method, we need the following notation. By the algebra of vector differentiation¹⁸ (Lütkepohl 2005, section A.13),

$$q \triangleq \frac{\partial \theta_h}{\partial \phi'} = \left(\sum_{j=0}^{h-1} \rho'^j, \sum_{j=0}^{h-1} \left[\sum_{i=0}^{j-1} (\beta' \rho^{j-1-i}) \otimes \rho'^i \right] \right), \quad (14)$$

$k \times (k+k^2)$

where $\otimes$ is the Kronecker product. Write $q = (q_1, q_2)$. Let $\hat{q} = (\hat{q}_1, \hat{q}_2)$, which is obtained by inserting $\hat{\beta}$ and $\hat{\rho}$ in (14).

The Wald test statistic is defined as

$$\mathcal{W}_{\text{IM}} = (R\hat{\theta}_h - R_0)' \left[R\hat{q} \left(\hat{\Omega} \otimes \left(\sum_{t=2}^n \tilde{x}_{t-1} \tilde{x}_{t-1}' \right)^{-1} \right) \hat{q}' R' \right]^{-1} (R\hat{\theta}_h - R_0), \quad (15)$$

where $\hat{\Omega} = n^{-1} \sum_{t=2}^n \hat{u}_t \hat{u}_t'$, with $\hat{u}_t = (\hat{u}_{0t}, \hat{u}_{xt})'$ being the OLS residual in (11) and (12). When $k=1$, the Wald statistic $\mathcal{W}_{\text{IM}}$ reduces to the square of $\mathcal{T}_{\text{IM}}(\theta_{h0})$, which was introduced in (10). Note that the calculation of $\mathcal{W}_{\text{IM}}$ only involves short-run OLS regressions (11) and (12).

As for the univariate predictive regression, the implementation of the Wald test based on (15) relies on a bootstrap procedure (see Appendix A for details).

5. Empirical Monte Carlos

5.1 The designs

We now evaluate the finite-sample performance via Monte Carlo experiments and compare a few long-horizon predictability tests. We first consider univariate predictive regression, and the pseudo-data are generated according to models (1) and (2). The main results are obtained for four data-generating processes (DGPs), DGPs A, B, C, and D. To maintain the empirical relevance of the designs, for each DGP, we use parameter values (in particular ρ and δ) that are all obtained by fitting the predictive regression using aggregate stock excess returns as the dependent variable and one of the predictors examined in the empirical section that follows (Section 6). These designs are chosen to cover different features of predictors and sampling frequencies of the data.

More specifically, DGP A mimics the predictive relationship between returns and log dividend-to-price ratios, the most widely used predictor, measured at annual frequencies ($n=92$) (Data set I, Section 6). This DGP is characterized by a predictor with strong persistence and strong negative endogeneity. It has an especially small sample size, as is common for annual data. DGPs B, C, and D mimic the predictive relationship using one of three other predictors, the Treasury-bill rate (TBL, quarterly observations, $n=264$, Data set II), the short interest index (SII, monthly observations, $n=504$, Data set III), or

¹⁸ The Online Appendix gives details about deriving Equation (14).

Table 1
Parameter values in Monte Carlo designs (DGPs A-D, univariate regression models (1) and (2))

Design	Fitting	n	ρ	δ	σ_0	σ_x	μ_y	μ_x
DGP A	DP, annual	92	0.898	-0.849	0.192	0.211	0.265	-0.355
DGP B	TBL, quarterly	264	0.953	-0.102	0.078	0.009	0.028	0.002
DGP C	SII, monthly	504	0.960	0.035	0.045	0.282	0.004	-0.009
DGP D	DFR, quarterly	264	-0.047	0.304	0.078	0.023	0.015	0.001

the default return spread (DFR, quarterly observations, $n=264$, Data set II), respectively; the first two predictors are characterized by strong persistence and low endogeneity, and the last one has low persistence. For completeness, Table 1 lists parameter values for DGPs A-D. The innovations u_t are i.i.d. bivariate Gaussian.

When generating pseudo-data, we use the same sample size as that of the corresponding data set in the empirical analysis for each DGP, and we examine over the same set of horizons, that is, up to $h=5$ for DGP A (annual data), up to $h=16$ for DGPs B and D (quarterly data), and up to $h=48$ for DGP C (monthly data). The initial value of x_t is set as the sample mean of the corresponding data series, and the first 500 observations in each realization are removed to reduce the initialization effect.

5.2 The tests

For the univariate predictive regression model, we consider five tests in the experiments. All tests are one sided. The first two are long-horizon-regression-based t -tests using the Newey-West standard error (with $h-1$ being the truncation point) and the standard error proposed by Hodrick (1992, section 1), both of which are popular in empirical studies. Both tests use standard normal critical values.

The next two are the implied t -test $\mathcal{T}_{IM}(0)$ and the long-horizon-regression-based scaled test, which are the focus of our asymptotic analysis. For the scaled test, we implement the test proposed by Hjalmarsen (2011), who uses a endogeneity correction and Bonferroni critical values, and we thus refer to it as Hjalmarsen’s test. Hjalmarsen’s test statistic is asymptotically normal, and thus should be compared with the one-sided $\mathcal{N}(0, 1)$ critical value (e.g., 1.645, if the test is right-tailed). We also implemented the test based on the t -statistic $\mathcal{T}_{Scaled}(0)$ without the endogeneity correction, using the robust confidence interval for c (Remark 4). We find that although it has better size control (in particular, under DGP D), but it is overall less powerful than Hjalmarsen’s test. Nevertheless, the difference is rather small and importantly, most properties of the scaled test, which we will discuss below, remain unchanged (Xu 2018).

For the proposed implied test, we use the bias-corrected bootstrap critical values, following Algorithm 1 and Remark A1 in Appendix A. In a working paper version (Xu 2018), we also implement the implied test with two other methods of obtaining critical values, one based on the Bonferroni procedure,

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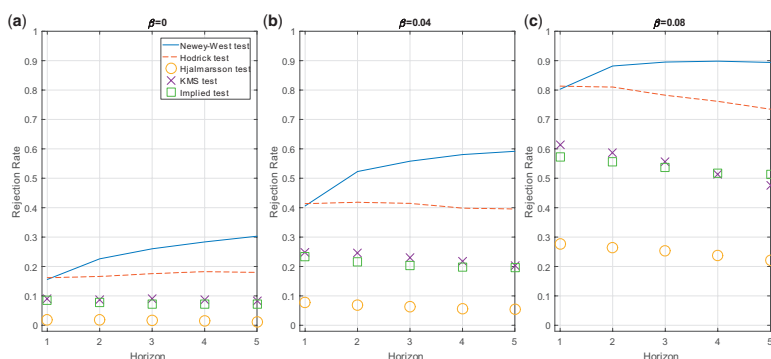


Figure 2
Finite-sample size and power of one-sided tests under DGP A ($\rho=0.898$, $\delta=-0.849$, $n=92$) against h

using the robust confidence interval for c (Remark 8), and the other based on a naive bootstrap procedure (without the bias correction), and find the performance of both tests is less competitive with the procedure we recommend here.¹⁹ The fifth test we consider is the one recently proposed by Kostakis et al. (2015, section 5) based on a quite different approach (using the instrumental variables regression, instead of OLS regression), which we refer to as the KMS test.

We evaluate the performance of various tests with respect to the null hypothesis $H_0: \theta_h=0$, and the one-sided alternative hypothesis (a left-tailed test for DGPs B and C, and a right-tailed test for DGPs A and D, motivated by the theory-implied sign of the predictive regression slope). For all the results reported in this section, the nominal size is 5%, and the number of replications is 5,000. The number of bootstrap replications is 1,000.

5.3 The results

Figures 2-9 report the main results on finite-sample rejection rates for various tests under different DGPs. There are two figures for each DGP. One figure plots the null rejection rate against the forecast horizon (e.g., Figure 2), for $\beta=0$ (reporting the size) and two other empirically relevant nonzero values of β (reporting the power), respectively, in three panels of the figure. The other figure plots the null rejection rate against β (e.g., Figure 3), for three horizons (horizon one, a moderate horizon and a long horizon) in three panels of the figure.

¹⁹ In particular, we find that the Bonferroni implied test is less powerful than the bootstrap implied test we implement here, and, nevertheless, it still offers substantial power improvement over the Hjalmarsson's scaled test. The naive bootstrap procedure has some obvious size distortion when the predictor is persistent and strongly (negatively) endogenous.

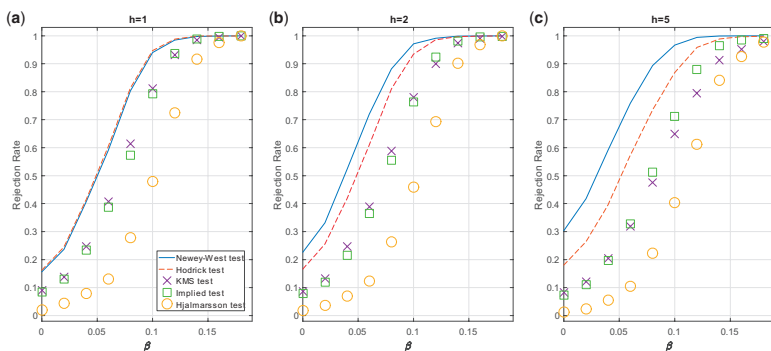


Figure 3
Finite-sample size and power of one-sided tests under DGP A ($\rho=0.898$, $\delta=-0.849$, $n=92$) against β

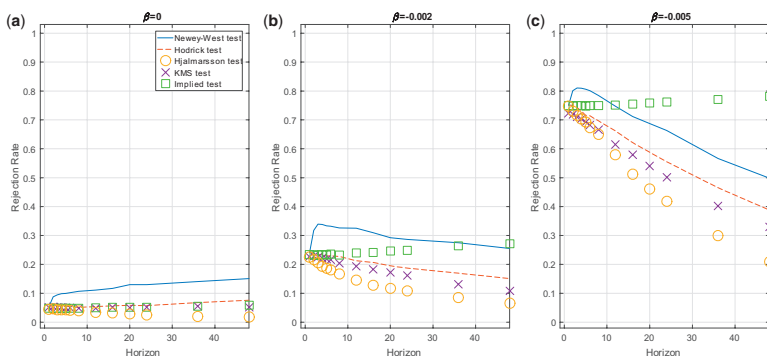


Figure 4
Finite-sample size and power of one-sided tests under DGP B ($\rho=0.953$, $\delta=-0.102$, $n=264$) against h

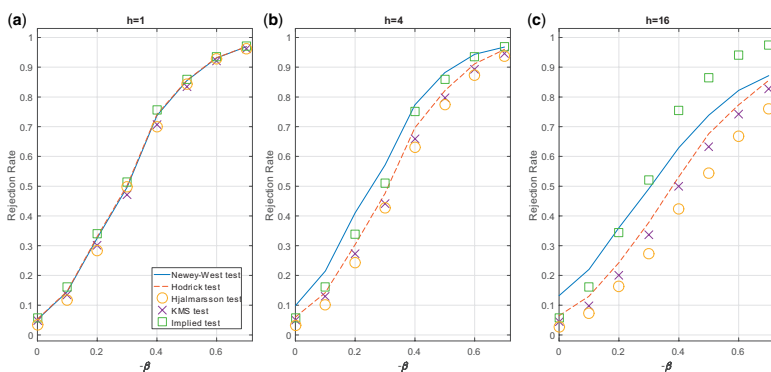


Figure 5
Finite-sample size and power of one-sided tests under DGP B ($\rho=0.953$, $\delta=-0.102$, $n=264$) against (the negative of) β

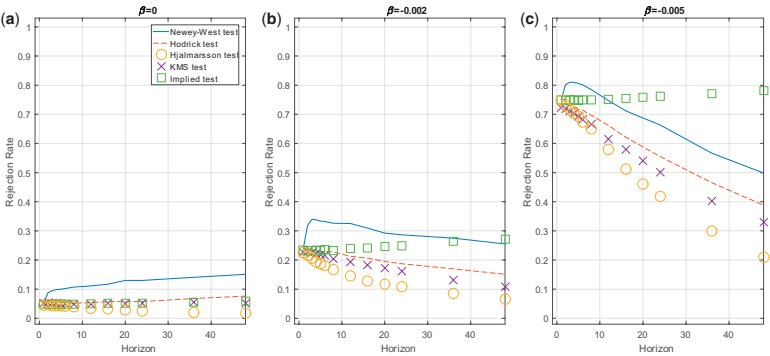


Figure 6
Finite-sample size and power of one-sided tests under DGP C ($\rho = 0.960$, $\delta = 0.035$, $n = 504$) against h

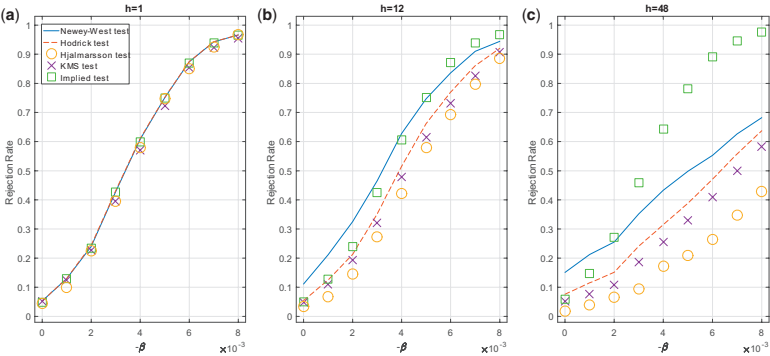


Figure 7
Finite-sample size and power of one-sided tests under DGP C ($\rho = 0.960$, $\delta = 0.035$, $n = 504$) against (the negative of) β

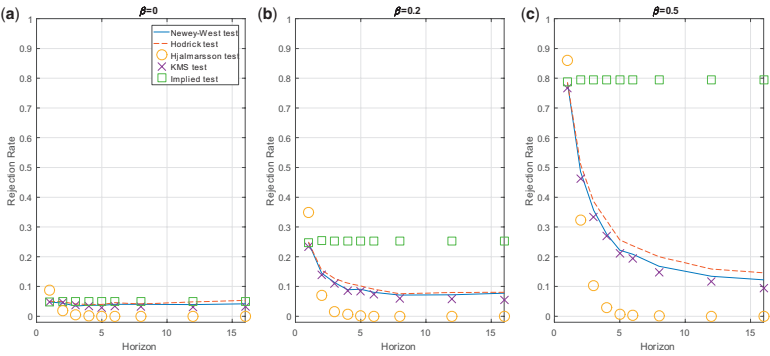


Figure 8
Finite-sample size and power of one-sided tests under DGP D ($\rho = -0.047$, $\delta = 0.304$, $n = 264$), against h

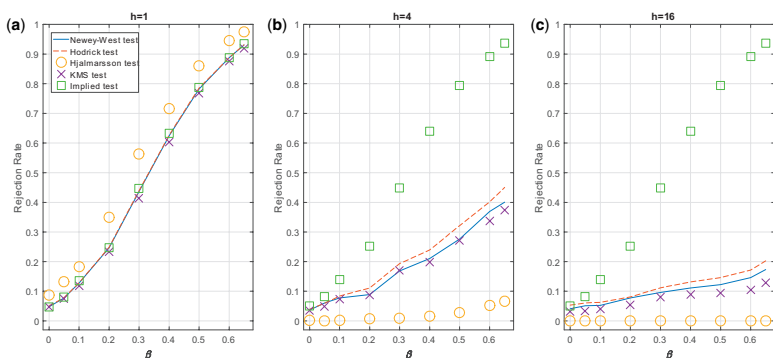


Figure 9
Finite-sample size and power of one-sided tests under DGP D ($\rho = -0.047$, $\delta = 0.304$, $n = 264$), against β

5.3.1 Newey-West and Hodrick tests. The Newey-West test works poorly under DGP A, which mimics the dynamics of returns and dividend-to-price ratios (Figure 2a). The empirical size (null rejection rate) is approximately 16%-30% for a test with nominal size 5% for horizons one to five, and the size distortion grows with the horizon. When $h = 1$, the Newey-West t -statistic reduces to the White t -statistic, and the overrejection is well understood from its non-Gaussian limit distribution (with some extension of the results in Campbell and Yogo (2006)), because of strong persistence of the predictor interacting with strong negative endogeneity. Figure 2a shows that the size distortion even worsens as the horizon increases.

For DGPs B and C, the Newey-West test has a size close to 5% when $h = 1$, since minor endogeneity almost makes the test asymptotically normal, and thus approximately valid even with strong persistence. The overrejection starts to pick up when the horizon increases, although it is much milder than that under DGP A, with the largest size about 15%. For DGP D, the Newey-West test has the size close to 5% for all horizons considered since the predictor has low persistence.

Why can the Newey-West test perform poorly over horizons? It is well known that the test overrejects seriously when serial correlation in $\tilde{x}_{t-1}u_{0t}(h)$ (estimated by $\tilde{x}_{t-1}\hat{u}_{0t}(h)$) is strong (Stock and Watson 2011). This exactly happens quickly when the horizon increases, as long as the predictor is not very weakly dependent (like DGP D). This happens even when $\tilde{x}_{t-1}\hat{u}_{0t}(h)$ is almost serially uncorrelated for $h = 1$, by construction. Figure 17 in the next section plots first-order autocorrelations of $\tilde{x}_{t-1}\hat{u}_{0t}(h)$ against h for the samples in the empirical analysis, which showcases this observation.

The Hodrick test, a favorite among empiricists (Hodrick 1992; Ang and Bekaert 2007; Bakshi, Panayotov, and Skoulakis 2011), coincides with the Newey-West test when $h = 1$. Under DGP A, it has obvious size distortion when $h = 1$, as the Newey-West test, but the size distortion does not increase

much over horizons. The empirical size is approximately between 15% and 19% over horizons, which still causes concerns when interpreting empirical results on the basis of this test. We also find that under this DGP, the distribution of the Hodrick test statistic is skewed to the right, so a two-sided test tends to obscure the difference and have less size distortion. The Hodrick test works well under DGPs B and C (low endogeneity) over horizons, with minor size distortion, and works almost perfectly under DGP D (low persistence).

As a result of lack of size control under the null, the Newey-West test also has higher rejection rates under the alternative hypothesis, than other tests considered. We include its power curve for completeness.

5.3.2 Hjalmarrsson's scaled test. We now discuss three more recent tests. Hjalmarrsson's scaled test maintains the size control below 5% for all horizons under DGPs A-C, but it's obviously conservative, in particular when endogeneity is strongly negative (Figure 2a, DGP A) or the horizon is long (Figure 6a, DGP C). This is consistent with the findings of the simulation study in Hjalmarrsson (2011), table 2, p. 828), which mostly focuses on persistent predictors.

Under DGP D, where the persistence of the predictor is low, Hjalmarrsson's test overrejects with a finite-sample size 8.8% when $h=1$. This is probably due to the incorrect coverage of the confidence interval for the AR slope based on inverting a unit root test, if the underlying process is not persistent, as pointed out by Phillips (2014). Strikingly, under this DGP D, the size falls to almost zero for any multiple horizon $h > 1$ (Figure 8a).

The undersized behavior of the scaled test we observe in most cases causes concerns for potential low power, which is confirmed in our analysis. This behavior originates from two sources. First, the Bonferroni procedure is conservative by nature. Second, more importantly, the scaling factor is too large when the predictor is not very persistent, which is a new finding in this paper. The effect is most obvious at a long horizon.

As a result, the scaled test has the lowest power among all tests considered in almost all cases (except for DGP D at $h=1$, where the test has incorrect size), in many cases by a quite large margin. For example, under DGP A, when $\beta=0.08$, the power of the scaled test is between 22.2% and 27.8% over all horizons, while the implied test proposed in this paper has power between 51.3% and 57.3% over horizons (Figure 2c). Under DGPs B and C, while the scaled test has similar or only slightly lower power compared with other tests at the first few horizons, there is obviously decreasing trend of the power when the horizon increases, and the power disadvantage is much enlarged at longer horizons. For DGP D, under which the scaled test has the most dismal performance, the power quickly drops to almost zero when the horizon is beyond the first few (Figures 8b and 8c). This confirms our prediction in the asymptotic analysis (Theorem 2); the test statistic itself converges to zero (thereby passing the

critical value at any significance level) if the persistence is dominated by the horizon.

We also consider the scaled test with the robust confidence interval for the AR slope, suggested by Mikusheva (2007) and Phillips (2014). The modified scaled test has a better size control regardless of the persistence level of the predictor, but it remains to have low power as Hjalmarrsson's test (Xu 2018).

5.3.3 Implied test and KMS test. The KMS test and the implied test perform well overall. Under DGP A, both tests have some size distortion (between 8.3% and 9% for KMS test, and between 7.2% and 8.5% for the implied test) over horizons, probably because of the relatively small sample size. Under other DGPs, the two tests have the size close to the nominal level 5% over all horizons considered.

Both KMS and implied tests have substantially higher power than the scaled test for all DGPs considered, across almost all multiple horizons. Furthermore, the implied test has certain degree of power advantage over the KMS test. Under DGP A, the KMS and implied tests have very close power. The power gain of the implied test over the KMS test exists noticeably for DGPs B, C, and D, in particular, at multiple horizons. For example, under DGP C (Figure 6), when $\beta = -0.002$, the power of the implied test increases from 23.4% to 27.1% from $h = 1$ to $h = 48$ (monthly observations), while the power of the KMS test decreases from 22.7% to 10.9%. For a farther alternative $\beta = -0.005$, the power of the implied test increases from 74.9% to 78.1% over horizons, while the power of the KMS test decreases from 72.4% to 32.9%. Under DGP D, the power of the KMS is very low and shrinks almost to the empirical size as the horizon increases, while the implied test maintains the power around 26% for $\beta = 0.2$ (Figure 8b) and around 80% for $\beta = 0.5$ (Figure 8c).

We notice that while the power curve against h has a monotonically decreasing pattern for the KMS test under all DGPs considered, the power curve is relatively stable for the implied test across horizons. Kostakis, Magdalinos, and Stamatogiannis (2015, section 5) obtain similar results in their simulations regarding the KMS test.

5.3.4 Robustness analysis. We now evaluate the performance of various tests if the DGP allows some serial correlation in the error of predictor dynamics u_{xt} . We consider four additional DGPs (denoted as DGPs A.ROB-D.ROB), similar to DGPs A-D, respectively, except that u_{xt} is allowed to be serially correlated, following an AR(1) model. The AR(1) slope of u_{xt} is empirically determined, using the data set underlying each of DGPs A-D. These slopes are small, taking values -0.056 , -0.102 , -0.157 , and 0.002 for DGPs A.ROB-D.ROB, respectively. Other parameters are also empirically determined, and they are slightly different from those for DGPs A-D. The Online Appendix gives details.

A general finding is that all tests have similar or slightly lower rejection rates under DGPs A.ROB-D.ROB than they do under DGPs A-D, under null and alternative hypotheses. All results regarding finite-sample behavior discussed above are qualitatively unchanged. Hjalmarsson's scaled test has substantially lower power than other tests considered in almost all cases. Both the implied test and KMS test remain to have good size control, and the implied test enjoys some power gain over the KMS test.

The largest error serial correlation occurs for DGP C.ROB. Both implied test and KMS test still have accurate size. The error serial correlation reduces the power of the implied test, and it does so for other tests as well. Comparing with an example of contrast mentioned above, under DGP C.ROB, when $\beta = -0.005$, the power of the implied test increases from 66.0% to 70.0% from $h = 1$ to $h = 48$ (monthly observations), while the power of the KMS test decreases from 64.0% to 25.4%; both tests have lower power than they have under DGP C. The performance of all tests under DGP D.ROB and DGP D are almost identical since the residual correlation is tiny. The Online Appendix provides the graphical results for the size and power of all tests under DGPs A.ROB-D.ROB.

5.4 Two predictors

We now consider the case with two predictors. The DGP follows models (11) and (12) with parameter values obtained by fitting the annual data on returns and two predictors DP and TBL we will use in Section 6.2.5. The sample size is $n = 92$. The parameter values are

$$\mu_y = 0.2912, \rho = \begin{pmatrix} 0.8979 & 0.1213 \\ 0.0004 & 0.9017 \end{pmatrix}, \Omega = \begin{pmatrix} 0.0361 & -0.0342 & 0.0002 \\ -0.0342 & 0.0443 & -0.0000 \\ 0.0002 & -0.0000 & 0.0002 \end{pmatrix}.$$

The coefficients in the horizon- h multiple predictive regression are denoted as θ_h^{DP} and θ_h^{TBL} for the two predictors, respectively. The tests in our experiments are Newey-West test, Hodrick test, KMS test, together with the implied test we proposed in Section 4 (which is implemented following Algorithm 1 in Appendix A). We didn't consider the scaled test as Hjalmarsson's test was proposed only for the univariate regression. The scaled test of this type requires a confidence region for the AR coefficients, so extending Hjalmarsson's test to allow for multiple predictors remains to be a major challenge computationally.

In this subsection, we are interested in three hypotheses regarding joint predictability, $H_0: \theta_h^{\text{DP}} = \theta_h^{\text{TBL}} = 0$, and individual predictability, $H_0: \theta_h^{\text{DP}} = 0$ and $H_0: \theta_h^{\text{TBL}} = 0$, against their two-sided alternatives. As before, Figures 10-14 report all results graphically. To evaluate the finite-sample size of various tests, we set $\beta = 0$ in (11). Figure 10 reports the results. To evaluate the power, we consider nonzero values $\beta = (0.1, -1.5)' \times s$ where s is a scale parameter between 0 and 1.2 controlling from the deviation from the null. The power of various tests as a function of horizon h , for a particular nonzero value $\beta = (0.1, -1.5)'$, is plotted in Figure 11 for three hypotheses. Figures 12-14

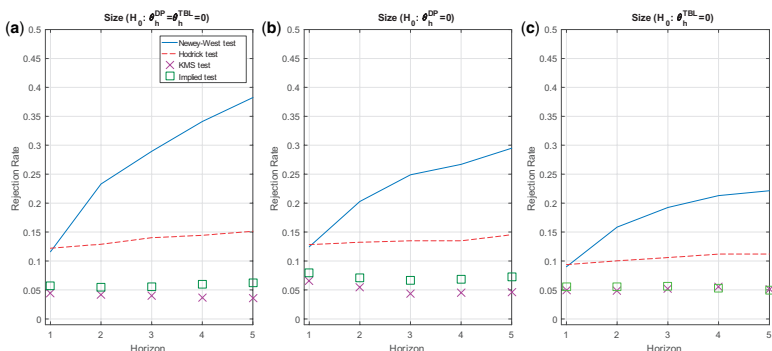


Figure 10
Finite-sample size of two-sided tests in predictive regression with two predictors
Panel a shows joint predictability, and panels b and c show individual predictability.

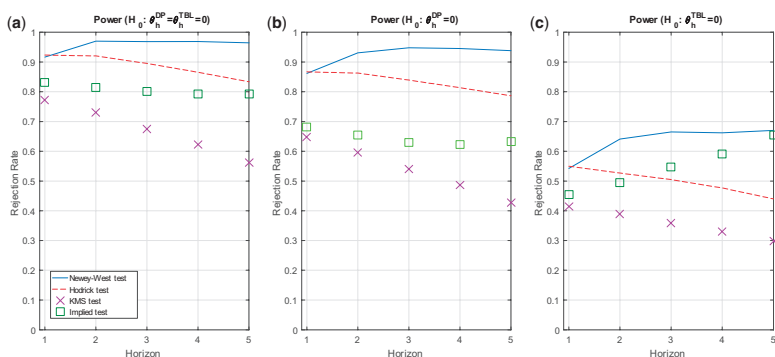


Figure 11
The power curve (against h) of two-sided test in predictive regression with two predictors, when $\beta = (0.1, -1.5)'$
Panel a shows joint predictability, and panels b and c show individual predictability.

show the power as a function of the deviation from the null (measured by s) for three hypotheses, respectively.

As from the figures, Newey-West test seriously overrejects and the size distortion deteriorates as the horizon increases, the phenomenon we have also seen in the univariate regression. The Hodrick test also overrejects, but the size distortion remains more stable over horizons. Both KMS test and the implied test have reasonable size, with some minor overrejections, probably for the small sample size. While the implied test is comparable to, or only slightly more powerful than the KMS test at short horizons, the power advantage of the implied test appears to be significant at longer horizons. Over three hypotheses, the largest power gain of the implied test over KMS is observed for $H_0: \theta_h^{\text{TBL}} = 0$.

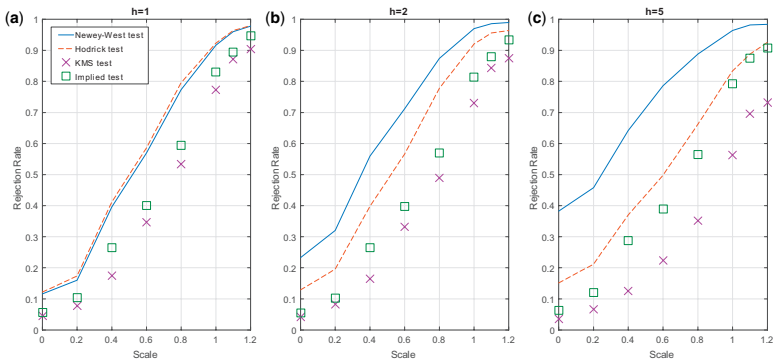


Figure 12
The power curve (against the alternative) of two-sided test in predictive regression with two predictors, when $\beta = (0.1, -1.5)' \times s$ with s being a scale parameter between $[0, 1.2]$ ($H_0: \theta_h^{\text{DP}} = \theta_h^{\text{TBL}} = 0$)

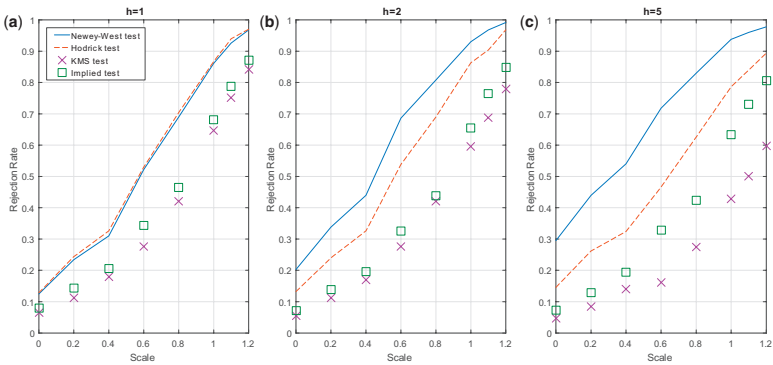


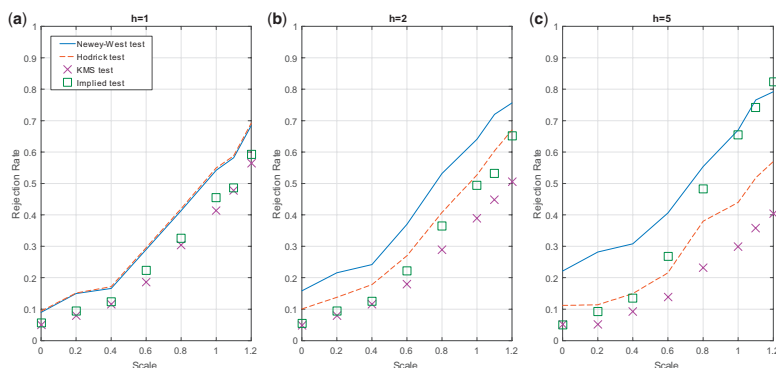
Figure 13
The power curve (against the alternative) of two-sided test in predictive regression with two predictors, when $\beta = (0.1, -1.5)' \times s$ with s being a scale parameter between $[0, 1.2]$ ($H_0: \theta_h^{\text{DP}} = 0$)

6. Empirical Illustration

We now empirically examine in-sample time-series predictability of excess returns in U.S. stock markets over various horizons using the popular predictors suggested in the literature and employing the proposed implication-based test and other existing tests.

6.1 The data

We use three data sets, each of which focuses on a particular predictor together with other commonly used predictors as competitors. The dependent variable is always the value-weighted market excess return. The first data set (Data set I) focuses on the predictor log dividend-to-price ratio (DP), using annual,

**Figure 14**

The power curve (against the alternative) of two-sided test in predictive regression with two predictors, when $\beta = (0.1, -1.5)' \times s$ with s being a scale parameter between $[0, 1.2]$ ($H_0: \theta_h^{\text{TBL}} = 0$)

quarterly, and monthly series spanning the years 1926–2017, the longest period available for all variables in the updated data set of Welch and Goyal (2008).²⁰

Data set II focuses on the predictor consumption-wealth ratio (CAY, Lettau and Ludvigson 2001), using quarterly series over 1952–2017, the longest period available for all variables in the updated data set of Welch and Goyal (2008). Other predictors include DP, dividend yield (DY), 3-month Treasury-bill rate (TBL), default return spread (DFR, the difference between long-term corporate bond and long-term government bond returns), and long-term returns (LTR, return on long-term government bonds).

Data set III focuses on a recently proposed predictor, the short interest index (SII, Rapach, Ringgenberg, and Zhou 2016), using monthly series spanning the years 1970–2014, the same sample as in Rapach, Ringgenberg, and Zhou (2016).²¹ The rationale behind the choice of the predictor is that “short sellers are informed traders who are able to anticipate changes in future aggregate cash flows and associated changes in future market returns” (Rapach, Ringgenberg, and Zhou 2016, p. 46). The short interest is calculated as the log of the equal-weighted mean of short interest (as a percentage of shares outstanding) across all publicly listed stocks on U.S. exchanges, and measures total short selling in the economy. The series is then detrended to form the short interest index (SII). Other predictors in this data set include DFR, LTR, term spread (TMS, long-term yield minus the Treasury-bill rate), DP, and DY. The definitions and details of these variables in Data sets I–III are available in Welch and Goyal (2008) and Rapach, Ringgenberg, and Zhou (2016).

To preview the empirical results, the analysis of Data sets II and III highlights the facts that the proposed implied test can produce meaningfully different

²⁰ Data source: Amit Goyal’s Web site (<http://www.hec.unil.ch/agoyal/>).

²¹ Data source: Dave Rapach’s Web site (<https://sites.google.com/slu.edu/daverapach>).

Table 2
Summary statistics (Data sets I and II)

Variable	Freq.	<i>n</i>	Mean	SD	1% quantile	Median	99% quantile
Welch-Goyal sample (1926-2017)							
Ex. ret.	Monthly	1,104	0.0051	0.0541	-0.1557	0.0094	0.1190
Ex. ret.	Quarterly	368	0.0154	0.1046	-0.3035	0.0303	0.2010
Ex. ret.	Annual	92	0.0620	0.1937	-0.5567	0.0903	0.4188
DP	Monthly	1,104	-3.3688	0.4610	-4.4241	-3.3390	-2.4292
DP	Quarterly	368	-3.3677	0.4640	-4.4592	-3.3457	-2.3949
DP	Annual	92	-3.3745	0.4689	-4.4434	-3.3509	-2.3818
Welch-Goyal sample (1952Q1-2017Q4)							
Ex. ret.	Quarterly	264	0.0153	0.0784	-0.2459	0.0274	0.1756
CAY	Quarterly	264	-0.0000	0.0239	-0.0502	0.0014	0.0417
DP	Quarterly	264	-3.5407	0.4045	-4.4747	-3.4885	-2.8152
DY	Quarterly	264	-3.5228	0.4052	-4.4796	-3.4746	-2.8023
TBL	Quarterly	264	0.0433	0.0311	0.0002	0.0430	0.1473
DFR	Quarterly	264	0.0009	0.0231	-0.0758	0.0012	0.0515
LTR	Quarterly	264	0.0166	0.0528	-0.0934	0.0098	0.2062

results from standard testing procedures, and that the scaled test can suffer from loss of power when the serial dependence of the predictor is weak or on the boundary of being classified as strongly persistent, as our asymptotic analysis suggests.

Table 2 reports the summary statistics for Data sets I and II. For the third data set, our calculations of summary statistics exactly replicate those reported by Rapach, Ringgenberg, and Zhou (2016, table 1, p. 50). We assess horizons up to $h=5, 16, 48$, for annual, quarterly, and monthly data, respectively.

For three data sets, Table 3 shows the key parameter estimates that would affect the performance of standard inference, in particular, the persistence of the predictor ρ and the endogeneity parameter δ . The table also gives the 95% confidence interval of the AR slope ρ using the robust method of Mikusheva (2007) and Phillips (2014), for each predictor series. It shows that persistence is clearly a feature for many predictors. The null hypothesis is return nonpredictability, and we use the alternative hypothesis of predictability on only one side of zero, for the results concerning univariate regression. This consideration is justified by the theory-implied sign of the slope; the slopes are negative for the predictors SII and TBL (e.g., more total short selling in the economy predicts lower future aggregate returns) and positive for all other predictors.

Figures 16, 19, 21, 23, and 25 report the testing results. For such a graph (e.g., Figure 16a), across forecast horizons, we plot the one-sided p -value for each test (except Hjalmarrsson's scaled test), which can be read from the left y -axis in the graph. For the scaled test, the construction of the test statistic depends on the significance level (when constructing the confidence interval for the AR slope), so we only plot the value of Hjalmarrsson's test statistic (or its negative, for TBL and SII) using the significance level 5%, read from the right y -axis in the graph. A value larger than 1.645 means a rejection by the scaled test.

Table 3
Preliminary data analysis for Data sets I-III

x_{t-1}	Freq.	n	$\hat{\rho}$	95% CI_{ρ}	$\hat{\delta}$	$\hat{\beta}$	R^2
Data set I: Welch-Goyal sample (1926-2017)							
DP	Monthly	1104	0.994	[0.989,1.004]	-0.977	0.0048	0.17%
DP	Quarterly	368	0.974	[0.956,1.010]	-0.950	0.0192	0.72%
DP (*, DGP A)	Annual	92	0.898	[0.830,1.039]	-0.849	0.0602	2.08%
Data set II: Welch-Goyal sample (1952Q1-2017Q4)							
CAY	Quarterly	264	0.971	[0.950,1.016]	-0.198	0.4594	1.94%
DP	Quarterly	264	0.977	[0.962,1.013]	-0.974	0.0186	0.92%
DY	Quarterly	264	0.978	[0.964,1.013]	-0.090	0.0218	1.27%
TBL (*, DGP B)	Quarterly	264	0.953	[0.925,1.010]	-0.102	-0.3009	1.42%
DFR (*, DGP D)	Quarterly	264	-0.047	[-0.156,0.077]	0.304	0.4714	1.93%
LTR	Quarterly	264	-0.040	[-0.148,0.084]	0.012	0.1407	0.90%
Data set III: Rapach-Ringgenberg-Zhou sample (1973M01-2014M12)							
SII (*, DGP C)	Monthly	504	0.960	[0.941,0.996]	0.035	-0.0050	1.23%
DFR	Monthly	504	-0.047	[-0.128,0.041]	0.231	0.3387	1.22%
LTR	Monthly	504	0.048	[-0.034,0.135]	0.112	0.1105	0.59%
TMS	Monthly	504	0.949	[0.929,0.985]	0.029	0.2191	0.54%
DP	Monthly	504	0.995	[0.991,1.008]	-0.989	0.0035	0.12%
DY	Monthly	504	0.995	[0.991,1.008]	-0.051	0.0040	0.15%

The outcome is always the value-weighted excess log return, and the predictor is x_{t-1} . The column “95% CI_{ρ} ” gives the 95% confidence interval of the AR slope ρ using the robust method of Mikusheva (2007) and Phillips (2014). Empirical Monte Carlo designs (DGPs A-D) mimic the dynamics of the series marked with (*).

In Figure 17, we also plot the first-order autocorrelations of $\tilde{x}_{t-1}\hat{u}_{0t}(h)$ for all samples. It is well know that the strong dependence in $\tilde{x}_{t-1}\hat{u}_{0t}(h)$ is the main driving force for the overrejections of the Newey-West test (Stock and Watson 2011). The figure shows that this symptom shows up quickly as the horizon increases (except for DFR and LTR in Data sets II and III, which are almost serially uncorrelated) even when the autocorrelation is minor at horizon one for all samples.

6.2 Empirical results

6.2.1 Data set I. Figures 15a-15c plot point estimates of the horizon- h slope of DP using two methods, one is based on the long-horizon regression and the other is based on implication. They are quite close at short horizons, but the implied estimates tend to be smaller than LH regression-based estimates at longer horizons. For example, for annual data, while they coincide at horizon one by construction ($\hat{\beta}=0.0602$), $\tilde{\theta}(h)$ and $\hat{\theta}(h)$ estimates are 0.141 and 0.114, respectively, at $h=2$, and 0.301 and 0.246, respectively, at $h=5$. Figure 15d shows R^2 s of long-horizon regressions for the data at three frequencies.

Figure 16 shows the results for various tests as functions of the horizon. The significance of the Newey-West t -test monotonically increases with the horizon, and the test is significant at the 5% level at most horizons, except the shortest few. This can be due to overrejections, based on our simulation results. Echoing prior research, as also shown in the figures here, the increasingly larger regression slopes, larger R^2 s, and more significant coefficient tests (based on the Newey-West standard error) over prediction horizons, reflect the

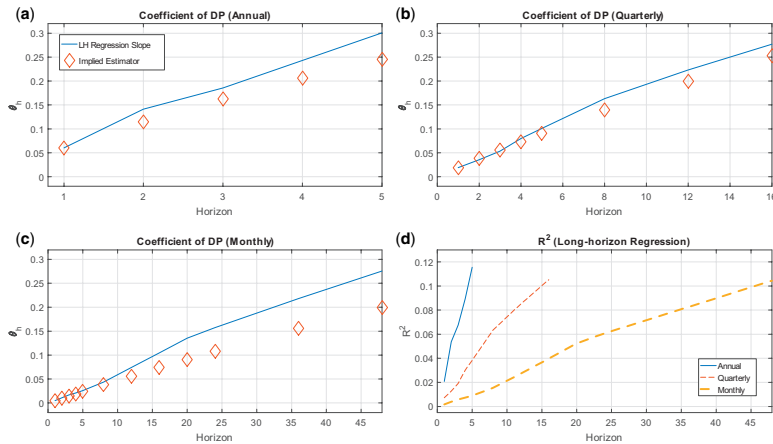


Figure 15
Point estimates of θ_h , when the predictor is the dividend-to-price (DP) ratio, using the annual, quarterly, and monthly data over 1926–2017

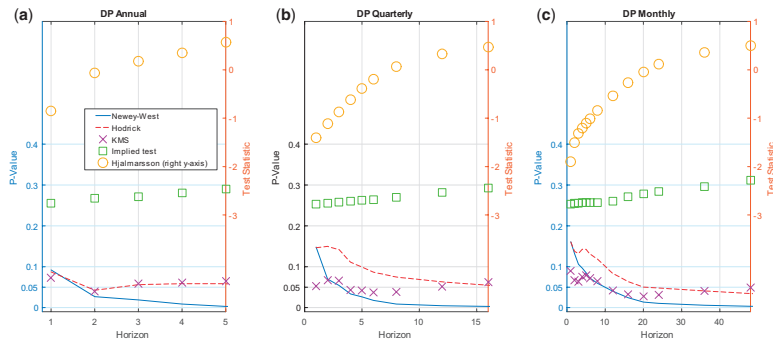


Figure 16
Various tests (one-sided) of multiple-horizon predictability, when the predictor is the dividend-to-price (DP) ratio, using the annual, quarterly and monthly data over 1926–2017
The figure shows Hjalmarsson’s test statistic at the 5% level (right y-axis) (thus the critical value is 1.645). For other tests, the figure shows the p -values (left y-axis).

folklore evidence that “stock returns are more predictable at longer horizons” (Campbell, Lo, and MacKinlay 1997; Cochrane 2005).

Other tests provide weaker evidence of significance at long horizons. Hodrick test is in general insignificant or marginally significant at the 5% level. The p -value of the KMS test is mostly floating around 5% across horizons. The implied test and Hjalmarsson’s scaled test are quite far from being significant for all horizons considered.

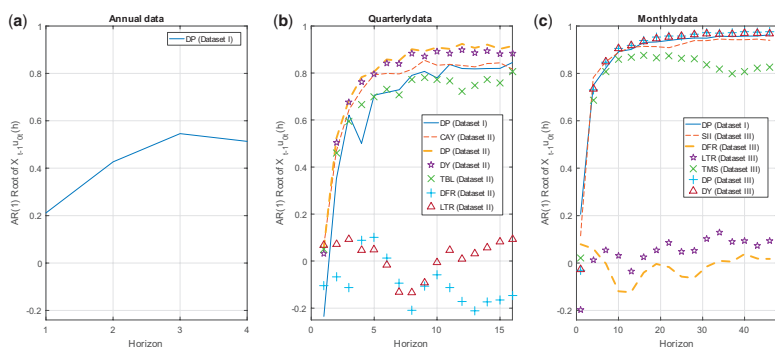


Figure 17
Autocorrelation of $\tilde{x}_{t-1}\hat{u}_{0t}(h)$ for all samples in Data sets I-III

6.2.2 Data set II. Now we turn to Data set II with quarterly observations. For 4 persistent predictors (CAY, DP, DY, and TBL), the pattern of slopes of LH regressions generally follows that of slopes estimated through implication (Figures 18 and 20), though two estimators tend to deviate from each other for longer horizons (in particular, for DY and TBL). For predictors with nearly no serial correlation (DFR and LTR), two estimators perform very differently; the implied estimators stabilize quickly as the horizon increases, and the slopes stay rather flat over the whole range of horizons. This is expected, as the added amount of predictability is very small over multiple horizons if the predictor itself is serially almost uncorrelated. In these cases (DFR and LTR), however, the LH regression slopes behave wildly across horizons (Figures 20b and 20c), probably because of the large estimation uncertainty induced by the large variance accumulated in the regression error $u_{0t}(h)$ across horizons (which does not occur for the implied estimator). Figures 22b and 22c depict the slopes of these two predictors using Data set III, and the similar observation follows.

The R^2 s of LH regressions (Figures 18d and 20d) increase with horizon for persistent predictors, but they largely decrease with horizon for predictors with almost no serial correlation. This corroborates our asymptotic analysis (Remark 1).

Regarding return predictable ability, CAY appears to strongly predict returns. All tests (except the scaled test) are significant at the 5% level at almost all horizons. LTR is also likely a strong predictor at all horizons. All tests (except the scaled test) are significant at 10% at almost all horizons. DP is not likely a strong predictor in this data set. The only evidence for predictability at the 5% level comes from a Newey-West test and perhaps a Hodrick test; however, that evidence may result from overrejections.

Th results for other predictors are less clear across all tests. One may attempt to believe TBL and DFR are strong predictors since the implied test at the 5% level is significant for both predictors. That the KMS test offers significance

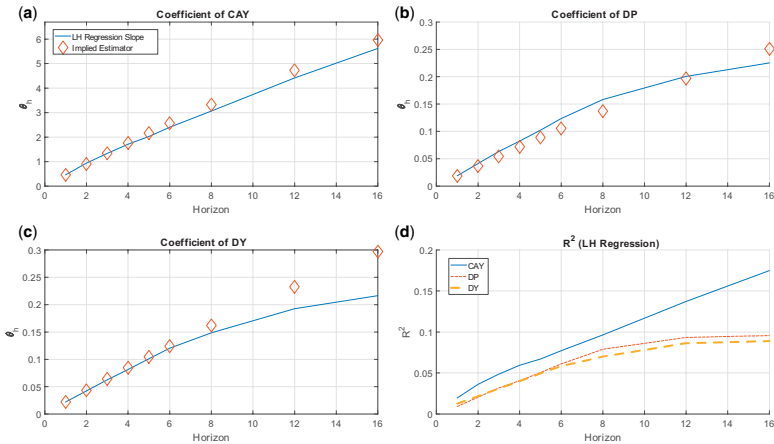


Figure 18
Point estimates of θ_h , when the predictor is CAY, DP, or DY, using the quarterly data over 1952–2017

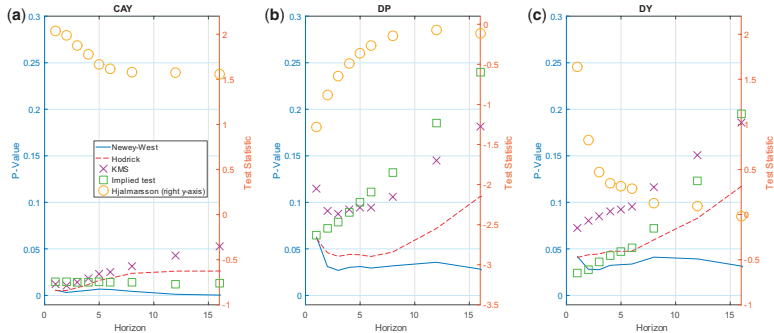


Figure 19
Various tests (one-sided) of multiple-horizon predictability, when the predictor is CAY, DP, or DY, using the quarterly data over 1952–2017
The figure shows Hjalmarsson's test statistic at the 5% level (right y-axis) (thus the critical value is 1.645). For other tests, the figure shows the p -values (left y-axis)

for TBL and DFR only at very short horizons, but not at other horizons, may attribute to its smaller power relative to the implied test at longer horizons.

The advantage of using the implied test is exemplified by the case of DFR. We note that for DFR, all tests considered except the implied test point to its inability to predict future returns for almost all multiple horizons (Figure 21b). This is likely due to the power advantage of the implied test as we have seen in Section 5 under DGP D. Figure 23b also illustrates a similar finding for the same predictor using Data set III.

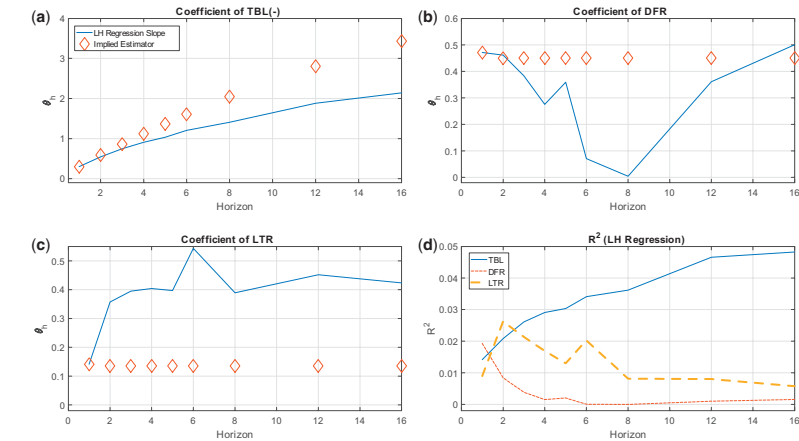


Figure 20
Point estimates of θ_h , when the predictor is TBL, DFR, or LTR, using the quarterly data over 1952–2017

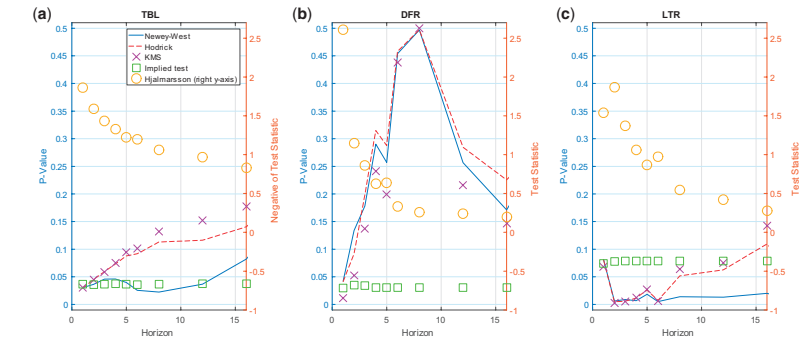


Figure 21
Various tests (one-sided) of multiple-horizon predictability, when the predictor is TBL, DFR, or LTR, using the quarterly data over 1952–2017
The figure shows (the negative of) Hjalmarsson's test statistic at the 5% level (right y-axis) (thus the critical value is 1.645). For other tests, the figure shows the p -values (left y-axis).

The scaled test generally provides very weak (if any) evidence for predictability, especially at multiple horizons. The test is far from significant at the 5% level for DP at all horizons but is significant or marginally significant for other predictors only at the first or second horizon. Significance is almost immediately gone as the horizon increases. We can identify a remarkable decaying trend in the scaled test statistic as the horizon increases for all predictors (DFR and LTR in particular) except DP. This is predicted by our asymptotic theory, which states that the scaled test statistic should converge to zero if the horizon becomes dominant over the persistence level of the predictor.

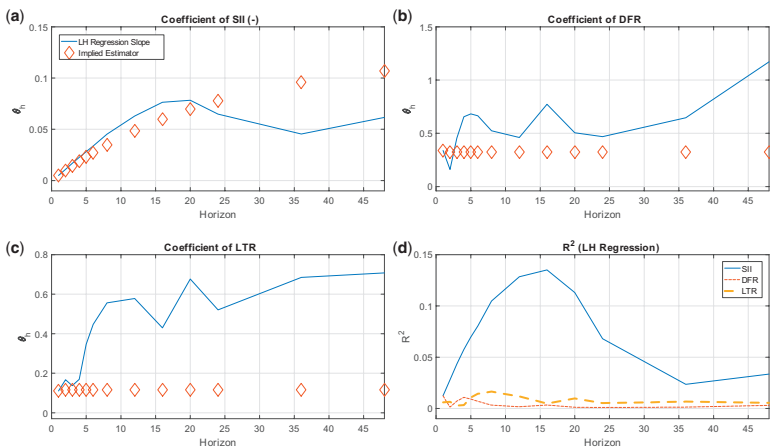


Figure 22
Point estimates of θ_h , when the predictor is (the negative of) SII, DFR, or LTR, using the monthly data over 1973–2014

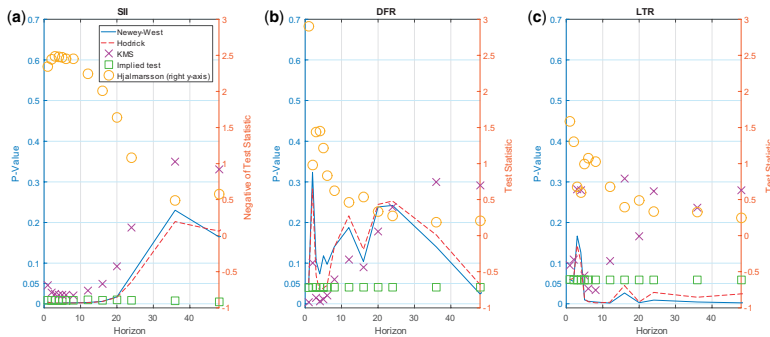


Figure 23
Various tests (one-sided) of multiple-horizon predictability, when the predictor is SII, DFR, or LTR, using the monthly data over 1973–2014

The figure shows (the negative of) Hjalmarsson’s test statistic at the 5% level (right y-axis) (thus the critical value is 1.645). For other tests, the figure shows the p -values (left y-axis).

6.2.3 Data set III. Using Data set III with monthly (and more recent) observations, the implied test proposed in this paper identifies (or confirms) four potentially useful predictors: SII, DFR, LTR, and TMS. The test shows that these predictors are all significant around the level 5% across all horizons. The finding that DFR is at the almost same level as SII in terms of predicting future returns over various horizons is missed by other tests. The KMS test detects predictability for these predictors at some horizons but misses it at other horizons. The traditional predictors DP and DY are not significant by all tests at almost all horizons.

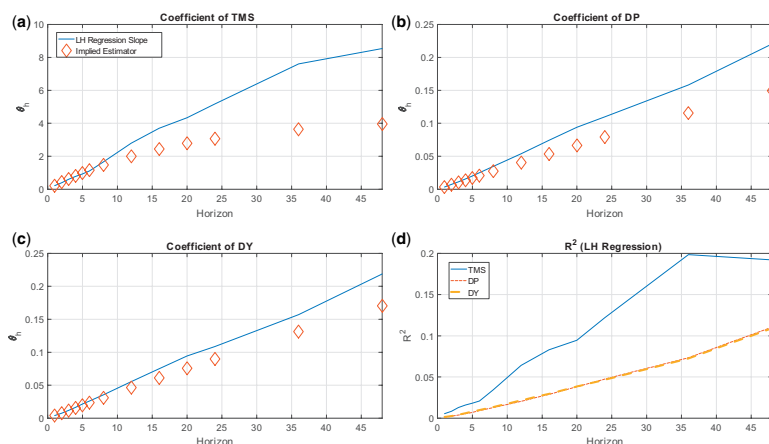


Figure 24
Point estimates of θ_h , when the predictor is TMS, DP, or DY, using the monthly data over 1973–2014

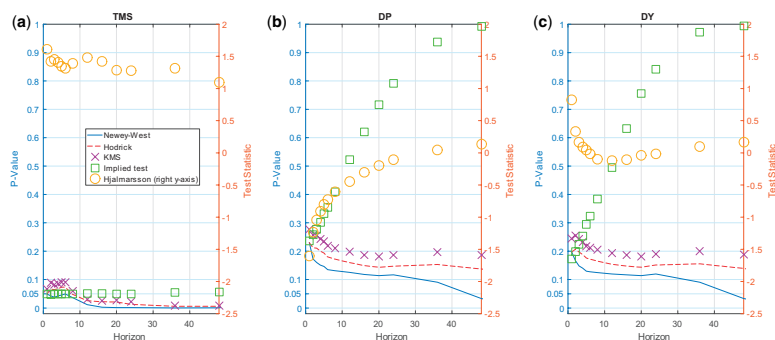


Figure 25
Various tests (one-sided) of multiple-horizon predictability, when the predictor is TMS, DP, or DY, using the monthly data over 1973–2014

The figure shows Hjalmarsson's test statistic at the 5% level (right y-axis) (thus the critical value is 1.645). For other tests, the figure shows the p -values (left y-axis)

Like what we have illustrated for the samples in Data set II, the scaled test still struggles to detect any predictability. The only evidence for significance we find in Data set III is for SII but only for horizons less than two years. For the predictors SII, DFR, and LTR, the scaled t -statistic quickly converges to zero as the horizon increases, which is just a mechanical artifact for this statistic if the predictor is not strongly persistent, instead of indicating lack of predictability.

6.2.4 Specification checking. The assumption of the AR(1) approximation of the dynamics of x_{t-1} works very well for all predictors considered in this section. In the Online Appendix, we report the autocorrelations of the

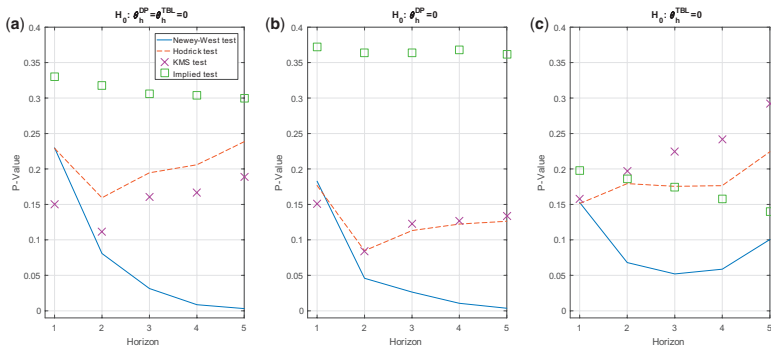


Figure 26
Various tests (two-sided) of multiple-horizon predictability, when the predictors are dividend-to-price (DP) ratio and Treasury-bill rate (TBL), using the annual data over 1926–2017
Panel a shows joint predictability, and panels b and c show individual predictability

predictor series and those of AR(1) residuals, providing support that most of the autocorrelation is captured by the first lags, and AR(1) residuals behave close to an uncorrelated sequence. In further misspecification analysis, we fit the AR(1) model to $\hat{u}_{xt}$ and find that the AR slope is minor for all series, with the largest one (in magnitude) being -0.157 , which occurs for the monthly SII series in Data set III. With robust standard errors, these slopes are insignificant at the 5% level except for the monthly SII series. Guided by the empirical Monte Carlo experiments under misspecification reported in Section 5.3.4, we conclude that such minor misspecification in most of our samples does not materially affect the validity of the implied test and its comparative advantage. The Online Appendix provides detailed results.

6.2.5 Two-predictor predictive regression. We now consider predictability tests using two predictors, dividend-to-price (DP) ratio and Treasury-bill rate (TBL), as suggested by Ang and Bekaert (2007). We use annual observations over years 1926–2017, as in Data set I. We focus on 4 two-sided tests, Newey–West, Hodrick, and KMS Wald tests as in Kostakis, Magdalinos, and Stamatogiannis (2015), which all use chi-squared critical values, together with the implied test suggested in Section 4, equipped with a bootstrap procedure as in Algorithm 1. The scaled test is mainly designed for univariate regression, and thus is not considered in this section.

Figure 26 reports the results for the joint and individual predictability tests. All tests, except for the Newey–West test, are insignificant at the 5% level at all horizons for all hypotheses. The Newey–West test is insignificant at short horizons, but becomes significant at longer horizons for the joint test and one individual predictability test. Guided by our simulation results in Section 5.4, which was designed to mimic the data in this case, it is highly probable that

Newey-West test leads to spurious rejection even when predictability does not exist.

6.2.6 Summary. Our results show that even with more powerful tests (against one-sided or two-sided alternatives), the widely used predictor the dividend-to-price ratio presents quite weak predictive ability for market returns when the persistence is properly accommodated, even at long horizons for which the strongest evidence typically occurs. Other predictors such as the consumption-wealth ratio, the short interest and the corporate-government bond returns spread provide stronger evidence than the dividend-to-price ratio, and the evidence is fairly coherent over various horizons.

7. Conclusion

Foreseeability of the price of stocks and bonds over longer (instead of shorter) periods was one of the key findings of the Laureates of the 2013 Nobel prize in economic sciences (Pressmeddelande 2013). Recent research on appropriate statistical inference procedures for long-horizon predictability has focused on accommodation of strong persistence in predictors. The current paper demonstrates a potential pitfall of the popular test based on the scaled t -statistic, which uses a too large standard error when the predictor is not persistent. The test based on the implication of the short-run model circumvents the necessity of dealing with the aggregated error, which is strongly dependent as the horizon increases. Unlike the scaled test, the implication-based test uses the persistence-adaptive standard error. Coupled with the bias-corrected bootstrap procedure we suggest, the implication-based test has attractive properties, as we elaborate in the paper, and is thus recommended.

The validity of the test hinges on specification of the autocorrelation structure of the predictor. Such specification has been proved fruitful in the literature on short-run predictability to design tests against unknown persistence of the predictor, and it underlies several popular robust procedures, prominently those of Campbell and Yogo (2006) and Phillips (2014). The alternative bias-correction approach (to short-run predictability), proposed by Stambaugh (1999), Amihud and Hurvich (2004), and Amihud et al. (2009), also relies on the correct specification of predictors' autocorrelation structure. Although we demonstrate that the level of mild misspecification in the empirical analysis is mostly harmless, an important direction for future work is to allow for flexible specification of the predictor dynamics.

Appendix

A. Implementation

ALGORITHM 1. (two-sided 100% Bootstrap Implied test for $H_0: R\theta_h = R_0$)

1. Run OLS regressions of (11) and (12) to obtain OLS estimates of coefficients $\{\hat{\mu}_y, \hat{\beta}, \hat{\mu}_x, \hat{\rho}\}$, and residuals $\hat{u}_t = (\hat{u}_{0t}, \hat{u}'_{xt})'$. Compute the Wald statistic $\mathcal{W}_{IM}$ as in (15).

- 2. Generate the bootstrap sample $\{y_t^*, x_t^* : t = 1, \dots, n\}$.
 - 2.1. Generate $u_t^* = \hat{u}_t e_t$, where e_t is a random number generated from the standard normal distribution. Write $u_t^* = (u_{0t}^*, u_{xt}^{*'})'$.
 - 2.2. Generate x_t^* and $y_t^* = \hat{\mu}_x + \hat{\rho}_{bc} x_{t-1}^* + u_{xt}^*$, where $\hat{\rho}_{bc}$ is the bias-corrected estimator of ρ , defined as

$$\hat{\rho}_{bc} = \hat{\rho} + \hat{\Omega}_x \left[(I_k - \hat{\rho}')^{-1} + \hat{\rho}' (I_k - \hat{\rho}^2)^{-1} + \sum_{j=1}^k \hat{\lambda}_j (I_k - \hat{\lambda}_j \hat{\rho}')^{-1} \right] \left(\sum_{t=2}^n \tilde{x}_{t-1} \tilde{x}_{t-1}' \right)^{-1}, \quad (A1)$$

where $\hat{\lambda}_j$'s ($j = 1, \dots, k$) are k eigenvalues of $\hat{\rho}'$. We set $x_1^* = x_1$.

- 2.3. Generate y_t^* as $y_t^* = \hat{\mu}_y + x_{t-1}^{*'} \hat{\beta} + u_{0t}^*$. We set $y_1^* = y_1$.
- 2.4. Repeat Steps 2.1-2.3 B times (e.g., $B = 1,000$).
- 3. For each bootstrap sample, calculate the bootstrap Wald statistic

$$\mathcal{W}_{IM}^* = (R\hat{\theta}_h^* - R\hat{\theta}_h)' \left[R\hat{q}^* \left(\hat{\Omega}^* \otimes \left(\sum_{t=2}^n \tilde{x}_{t-1}^* \tilde{x}_{t-1}^{*'} \right)^{-1} \right) \hat{q}^{*'} R' \right]^{-1} (R\hat{\theta}_h^* - R\hat{\theta}_h),$$

where $\hat{\theta}_h^*, \hat{q}^*, \hat{\Omega}^*$ are computed like $\hat{\theta}_h, \hat{q}, \hat{\Omega}$ except that the bootstrap sample $\{y_t^*, x_t^*\}$ in place of the original sample $\{y_t, x_t\}$.

- 4. Use the $100(1-\epsilon)\%$ quantile $Q_{\mathcal{W}_{IM}^*}(1-\epsilon)$ (e.g., $\epsilon = 0.05$) of $\mathcal{W}_{IM}^*$ over B bootstrap replications as the critical value, that is, the hypothesis H_0 is rejected at the level ϵ if $\mathcal{W}_{IM} > Q_{\mathcal{W}_{IM}^*}(1-\epsilon)$. Here, $Q_X(\epsilon)$ denotes the $100\epsilon\%$ quantile of the random variable X . □

Remark A1. In the univariate predictive regression, the bootstrap procedure above can be similarly applied to the t -statistic $\mathcal{T}_{IM}(0)$. We can also form a one-sided test if we know the sign of the slope θ_h , for example, via economic theory. To implement it, supposing that the sign is known to be positive (e.g., the slope of the dividend-to-price ratio), we can use the $100(1-\epsilon)\%$ quantile $Q_{\mathcal{T}_{IM}^*(0)}(1-\epsilon)$ of $\mathcal{T}_{IM}^*(0)$ as the critical value; that is, the hypothesis $H_0 : \theta_h = 0$ against $H_A : \theta_h > 0$ is rejected at the level ϵ if $\mathcal{T}_{IM}(0) > Q_{\mathcal{T}_{IM}^*(0)}(1-\epsilon)$. Similarly, if the sign is known to be negative (e.g., the slope of Treasury-bill rates), the hypothesis $H_0 : \theta_h = 0$ against $H_A : \theta_h < 0$ is rejected at the level ϵ if $\mathcal{T}_{IM}(0) < Q_{\mathcal{T}_{IM}^*(0)}(\epsilon)$. In the univariate predictive regression, the bias-corrected estimator of ρ can be computed as $\hat{\rho}_{bc} = \hat{\rho} + (n-1)^{-1}(1+3\hat{\rho})$ (Stambaugh 1999), consistent with (A1), or to a higher order as $\hat{\rho}_{bc2} = \hat{\rho} + (n-1)^{-1}(1+3\hat{\rho}) + (n-1)^{-2}3(1+3\hat{\rho})$ (Amihud and Hurvich 2004).

Remark A2. The bias expression (A1) comes from Nicholls and Pope (1988); see also Amihud et al. (2009). The naive bootstrap uses $\hat{\rho}$ (instead of $\hat{\rho}_{bc}$) in Step 2.2. For the univariate predictive regression model, it is possible to prove that the implied statistic using the naive bootstrap is distributionally valid for $0 \leq r < 1$ under the null hypothesis. However, when $r = 1$, the naive bootstrap is not consistent in that the bootstrap implied statistic has a limit distribution $\tau_1(\hat{c}, \delta)$ conditional on the data, where $\hat{c} = n(\hat{\rho} - 1)$. Using the unbiased estimator of ρ (to the first or higher order) in generating the bootstrap data, as the biased-corrected bootstrap does, helps to mitigate the inconsistency issue. Our empirical Monte Carlos in Section 5 show that the method works very well in finite samples.

Remark A3. The residual-based wild bootstrap (Wu 1986; Liu 1988) employed in Step 2.1 is useful to recover the null distribution even when the error is heteroscedastic. This is in contrast to the standard bootstrap that takes random draws from $\{\hat{u}_t\}$.

B. Asymptotic Theory for Long-Horizon Regression Estimator and Implied Estimator

We will now present the asymptotic distributions of $\tilde{\theta}_h$ and $\hat{\theta}_h$ in Theorems A1 and A2, respectively.

Theorem A1 (Asymptotics for the LH-regression-based slope estimator $\tilde{\theta}_h$). Suppose Assumption 1 holds.

(i) Suppose $\beta=0$. As $n \rightarrow \infty$,

$$\begin{cases} nh^{-1}\tilde{\theta}_h \xrightarrow{d} \sigma_x^{-1}\sigma_0 \left(\int \tilde{J}_{x,c}^2\right)^{-1/2} \tau_1(c, \delta), & \text{if } r=1; \\ n^{(1+r)/2}h^{-1}\tilde{\theta}_h \xrightarrow{d} \mathcal{N}(0, (-2c)\sigma_x^{-2}\sigma_0^2), & \text{if } 0 < r_h < r < 1; \\ n^{1/2}h^{-1/2}\tilde{\theta}_h \xrightarrow{d} \mathcal{N}(0, 4\sigma_0^2\sigma_x^{-2}), & \text{if } r_h > r > 0; \\ n^{1/2}h^{-1/2}\tilde{\theta}_h \xrightarrow{d} \mathcal{N}(0, (3-\rho)(1+\rho)\sigma_0^2\sigma_x^{-2}), & \text{if } r_h > r=0. \end{cases}$$

(ii) Suppose $\beta \neq 0$. As $n \rightarrow \infty$,

$$\begin{cases} nh^{-2}(\tilde{\theta}_h - \theta_h) \xrightarrow{d} \beta \left(\int \tilde{J}_{x,c}^2\right)^{-1/2} \tau_2(c)/2, & \text{if } r=1; \\ n^{(1+r)/2}h^{-2}(\tilde{\theta}_h - \theta_h) \xrightarrow{d} \mathcal{N}(0, (-c)\beta^2/2), & \text{if } 0 < r_h < r < 1; \\ n^{1/2-r-\epsilon}h^{-1/2}(\tilde{\theta}_h - \theta_h) \xrightarrow{p} 0, & \text{if } r_h > r \geq 0. \end{cases}$$

Theorem A2 (Asymptotics for the implied estimator $\hat{\theta}_h$). Suppose Assumption 1 holds.

(i) Suppose $\beta=0$. As $n \rightarrow \infty$,

$$\begin{cases} nh^{-1}\hat{\theta}_h \xrightarrow{d} \sigma_x^{-1}\sigma_0 \left(\int \tilde{J}_{x,c}^2\right)^{-1/2} \tau_1(c, \delta), & \text{if } r=1; \\ n^{(1+r)/2}h^{-1}\hat{\theta}_h \xrightarrow{d} \mathcal{N}(0, (-2c)\sigma_x^{-2}\sigma_0^2), & \text{if } 0 < r_h < r < 1; \\ n^{(1-r)/2}\hat{\theta}_h \xrightarrow{d} \mathcal{N}(0, 2(-c)^{-1}\sigma_x^{-2}\sigma_0^2), & \text{if } r_h > r \geq 0; \\ n^{1/2}\hat{\theta}_h \xrightarrow{d} \mathcal{N}(0, (1+\rho)(1-\rho)^{-1}\sigma_x^{-2}\sigma_0^2), & \text{if } r_h > r=0. \end{cases}$$

(ii) Suppose $\beta \neq 0$. As $n \rightarrow \infty$,

$$\begin{cases} nh^{-2}(\hat{\theta}_h - \theta_h) \xrightarrow{d} \beta \left(\int \tilde{J}_{x,c}^2\right)^{-1/2} \tau_2(c)/2, & \text{if } r=1; \\ n^{(1+r)/2}h^{-2}(\hat{\theta}_h - \theta_h) \xrightarrow{d} \mathcal{N}(0, (-c)\beta^2/2), & \text{if } 0 < r_h < r < 1; \\ n^{(1-3r)/2}(\hat{\theta}_h - \theta_h) \xrightarrow{d} \mathcal{N}(0, 2(-c)^{-3}\beta^2), & \text{if } r_h > r > 0; \\ n^{1/2}(\hat{\theta}_h - \theta_h) \xrightarrow{d} \mathcal{N}\left(0, \frac{1+\rho}{1-\rho} \left[\sigma_0^2\sigma_x^{-2} + \frac{2\beta\sigma_{0x}}{(1-\rho)\sigma_x^2} + \frac{\beta^2}{(1-\rho)^2} \right] \right), & \text{if } r_h > r=0. \end{cases}$$

C. Proofs of Theorems

The Online Appendix provides the proofs of Theorems 1-4 and Theorems A1 and A2. Preliminary results are summarized in the following Lemmas 1-3, the proofs of which are also provided in the Online Appendix.²²

For a (non)random quantity ξ , we write $\xi = \xi_1 + o_p(\xi_1)$ simply as $\xi = \xi_1 + \text{s.o.}$, where s.o. means a term of smaller order in probability.

Lemma 1. Under the assumptions in (6) and (7), the following results hold.

(i) For θ_h ,

$$q_1 = \sum_{j=0}^{h-1} \rho^j = \begin{cases} h + \text{s.o.}, & \text{if } r_h < r; \\ (-c)^{-1}n^r + \text{s.o.}, & \text{if } r_h > r \geq 0, \end{cases} \quad (\text{A2})$$

and

$$q_2 = \sum_{j=0}^{h-1} j\rho^{j-1}\beta = \begin{cases} h(h-1)\beta/2 + \text{s.o.}, & \text{if } r_h < r; \\ c^{-2}n^{2r}\beta + \text{s.o.}, & \text{if } r_h > r \geq 0. \end{cases}$$

²² <https://sites.google.com/site/xukeli2015/research>.

(ii) For $\tilde{\theta}_h$, recall that $\varphi_j = \sum_{i=0}^{h-2-j} \rho^i$, then

$$\sum_{j=0}^{h-2} \varphi_j = \begin{cases} h(h-1)/2 + \text{s.o.}, & \text{if } r_h < r; \\ (-c)^{-1} n^r h + \text{s.o.}, & \text{if } r_h > r \geq 0, \end{cases}$$

$$\sum_{j=0}^{h-2} \varphi_j^2 = \begin{cases} h^3/3 + \text{s.o.}, & \text{if } r_h < r; \\ c^{-2} n^{2r} h + \text{s.o.}, & \text{if } r_h > r \geq 0, \end{cases} \quad (\text{A3})$$

and

$$\sum_{j=0}^{h-2} \varphi_j^2 + 2 \sum_{k=1}^{h-2} \rho^k \sum_{j=k}^{h-2} \varphi_j \varphi_{j-k} = \begin{cases} h^4/4 + \text{s.o.}, & \text{if } r_h < r; \\ 2(-c)^{-3} n^{3r} h + \text{s.o.}, & \text{if } r_h > r > 0; \\ h(1-\rho)^{-3}(1+\rho) + \text{s.o.}, & \text{if } r_h > r = 0. \end{cases}$$

Lemma 2. Let $u_{0t}(h)$ be as defined in (5), which takes the MA($h-1$) form of

$$u_{0t}(h) = \sum_{j=0}^{h-1} \bar{\Phi}'_j u_{t+j},$$

where $\bar{\Phi}_j = (1, \bar{\varphi}_j \beta)'$ with $\bar{\varphi}_j = \varphi_j$ (for $0 \leq j \leq h-2$) and $\bar{\varphi}_{h-1} = 0$. Suppose $0 \leq r < 1$. Denote $V_{lr}^u = \text{Var}(n^{-1/2} \sum_{t=2}^n u_{0t}(h))$ and $V_{lr} = \text{Var}(n^{-1/2} \sum_{t=2}^n x_{t-1} u_{0t}(h))$, where,

$$V_{lr} = \frac{\sigma_x^2}{1-\rho^2} \left(\sum_{j=0}^{h-1} \bar{\Phi}'_j \Omega \bar{\Phi}_j + 2 \sum_{k=1}^{h-1} (1-k/n) \rho^k \sum_{j=k}^{h-1} \bar{\Phi}'_j \Omega \bar{\Phi}_{j-k} \right),$$

Under Assumption 1, asymptotics of V_{lr} and V_{lr}^u are given below.

(i) If $\beta = 0$, then $h^{-2} V_{lr}^u \rightarrow \sigma_0^2$, and

$$\begin{cases} h^{-2} n^{-r} V_{lr} \rightarrow (-2c)^{-1} \sigma_x^2 \sigma_0^2, & \text{if } 0 < r_h < r < 1; \\ h^{-1} n^{-2r} V_{lr} \rightarrow c^{-2} \sigma_x^2 \sigma_0^2, & \text{if } r_h > r > 0; \\ h^{-1} V_{lr} \rightarrow (1-\rho)^{-2} \sigma_x^2 \sigma_0^2, & \text{if } r = 0. \end{cases}$$

(ii) If $\beta \neq 0$, then

$$\begin{cases} h^{-4} V_{lr}^u \rightarrow \beta^2 \sigma_x^2 / 4, & \text{if } r_h < r; \\ n^{-2r} h^{-2} V_{lr}^u \rightarrow \beta^2 \sigma_x^2 c^{-2}, & \text{if } r_h > r > 0; \\ h^{-2} V_{lr}^u \rightarrow \sigma_0^2 + \beta \frac{2\sigma_0 x}{1-\rho} + \frac{\beta^2 \sigma_x^2}{(1-\rho)^2}, & \text{if } r_h > r = 0. \end{cases}$$

and

$$\begin{cases} h^{-4} n^{-r} V_{lr} \rightarrow (-c)^{-1} \sigma_x^4 \beta^2 / 8, & \text{if } 0 < r_h < r < 1; \\ n^{-4r} h^{-1} V_{lr} \rightarrow (-c)^{-4} \sigma_x^4 \beta^2, & \text{if } r_h > r > 0; \\ h^{-1} V_{lr} \rightarrow \frac{\sigma_x^2 \sigma_0^2}{(1-\rho)^2} + \frac{2(1+\rho^2) \sigma_x^2 \sigma_0 x}{(1-\rho)^4 (1+\rho)} \beta + \frac{\sigma_x^4}{(1-\rho)^4} \beta^2, & \text{if } r_h > r = 0. \end{cases}$$

Lemma 3. Suppose Assumption 1 holds. If $\beta = 0$,

$$n^{-1} h^{-1} \sum_{t=2}^n \hat{u}_{0t}^2(h) \xrightarrow{p} \sigma_0^2.$$

If $\beta \neq 0$,

$$\begin{cases} n^{-1} h^{-3} \sum_{t=2}^n \hat{u}_{0t}^2(h) \xrightarrow{p} 3^{-1} \beta^2 \sigma_x^2, & \text{if } 0 < r_h < r \leq 1; \\ n^{-(1+2r)} h^{-1} \sum_{t=2}^n \hat{u}_{0t}^2(h) \xrightarrow{p} c^{-2} \beta^2 \sigma_x^2, & \text{if } r_h > r > 0; \\ n^{-1} h^{-1} \sum_{t=2}^n \hat{u}_{0t}^2(h) \xrightarrow{p} \sigma_0^2 + \frac{2\sigma_0 x \beta}{1-\rho} + \frac{\sigma_x^2 \beta^2}{(1-\rho)^2}, & \text{if } r = 0. \end{cases}$$

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